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An Agent-Based Approach to Simulating Systemic Shifts

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1 Introduction

The accelerating climate crisis poses a systemic challenge that calls for systemic responses. As scientific consensus on the need for rapid greenhouse-gas reduction has strengthened, policymakers have turned increasingly to market-based instruments such as emissions-trading systems (ETS) and carbon taxes. The European Union Emissions Trading Scheme (EU ETS), launched in 2005, remains the most comprehensive cap-and-trade programme in operation.

Yet the empirical record of such instruments is mixed. Some researchers have questioned their ability to deliver deep decarbonisation based on ex-post analyses, noting e.g. susceptibility to low prices, allowance oversupply and management costs due to complexity [Green, 2021]. Concerns suggest that effective climate policy must address not only emissions but also the structural, distributive and institutional dimensions of economic transition [Green and Denniss, 2021].

In parallel, a growing body of literature argues that a government acting as an employer of last resort can guarantee full employment [Fullwiler, 2007], which could be a promising approach to steer production towards green infrastructure while simultaneously improving social outcomes: A Job-Guarantee programme could convert involuntary unemployment into paid work on projects with clear social or ecological value, stabilizing aggregate demand and supporting households during structural change.

This thesis therefore asks whether such an employment-driven policy alternative can be modeled within a stock-flow-consistent, agent-based macroeconomic (ABM) framework. Rather than delivering a finished simulation, the work focuses on the design of an architecture capable of capturing the interplay among energy flows, emissions, monetary circuits and institutional interventions.

The constructed model economy consists of heterogeneous households, firms, commercial banks, a central bank and a government. Energy acts as an explicit production constraint, emissions arise endogenously and the financial system creates credit and funds endogenously while distributing liquidity. Due to the quickly growing complexity of the code a stylized carbon-tax-and-rebate scheme stands in for a full ETS for now, and a configurable Job-Guarantee programme provides the employer-of-last-resort functionality.

Accordingly, the work should be read as a model prototype: it outlines architecture, current limitations and avenues for extension. Its main contribution lies in integrating ecological constraints, public-employment mechanisms and agent-based techniques within a single simulation environment. Although complete numerical results have to remain outside the present scope, the design demonstrates the feasibility of constructing an integrated 'system of systems' simulation platform in which macroeconomic patterns emerge from the interaction of physical limits, institutional structure, policy choices and individual behavior. By doing so, this thesis aims to lay a conceptual foundation for building a more elaborate platform. The model aimed for would allow for studying macroeconomic dynamics in the context of ecological constraints, institutional design, and monetary complexity simultaneously while capturing interactions that might be missed otherwise when studying complex systems. Tools like that could become increasingly necessary as nations have to navigate an era of systemic transformations.

2 Conceptual Overview

This section presents the structural and conceptual core of the macroeconomic ABM developed in this thesis. Following the motivation laid out in the introduction, the core macroeconomic questions that guide the design include: How does a carbon price reshape sectoral demand and production structure? Can a job guarantee policy support full employment without inflationary pressure while aiding in the transition to a non fossil fuel dependent economy? Could the financial sector act as an indirect production constraint in an otherwise physically feasible system? Which

distributional effects arise under different transition strategies?

The developed model currently simulates a closed monetary economy consisting of production, consumption, government intervention, banking, and energy systems. The architecture shares a stock-flow-consistent backbone [à la Godley and Lavoie, 2007] and some key structural elements in terms of agent taxonomy with prior work [Caiani et al., 2015], extending it with an approach that is able to model physical good properties and production chains. Additionally it prioritizes extensibility with each subsystem implemented as an independent component to support iterative development. This separation of concerns allows the system to evolve without directly compromising the overall simulation structure or its underlying financial logic. The simulation itself iterates through a step-based loop, each step representing a month passing, sequentially executing phases associated with production, market exchange, policy enforcement, and financial settlement along the way. Migrating from this sequential to a more dynamic asynchronous or event-driven approach is left for a future iteration due to increased programmatic complexity. Migrating the model from its current object-oriented design to an entity-component-system would likely be necessary. Key architectural principles include:

Stock-flow consistency: All financial flows are fully tracked across sectoral balance sheets, ensuring double-entry coherence at each step.

Physical realism: Goods are modeled as discrete, physical entities with specific characteristics. Energy flows and emissions arise endogenously from production and infrastructure constraints, anchoring the system in material throughput.

Agent-based flexibility: Decision-making and behavioral routines are embedded at the agent level, enabling local adaptation, heterogeneity, and path dependence.

Policy modularity: Both carbon pricing and direct public employment programs can be activated, extended, or combined to test complementary or competing institutional configurations.

Particular emphasis is placed on structural transformation scenarios, including a decarbonisation pathway and a full employment strategy. The currently developed structure allows for the following policy experiments: (i) a baseline scenario without intervention, (ii) a progressively increasing carbon tax with household rebates, (iii) a public job guarantee program coordinated through government-led clean energy investment, and (iv) a joint scenario combining both instruments. These policies operate within a broader macro-financial framework, enabling simulations of gradual shifts in energy usage within the system as well as extended effects of policy interventions, including changes in wealth distribution, employment levels, and aggregate output. Utilizing globally set seed values for stochastically determined or augmented parameters within the model, the architecture allows building randomized worlds while making policy-based interventions comparable against the baseline scenario.

The models modular structure supports the gradual addition of new agents, goods, institutions, and behavioral routines. While current performance is sufficient for small-scale simulation and will support additional extensions if handled correctly, optimization and scaling remain long-term goals. A future migration from the current Python-based implementation to a compiled backend (e.g. Rust or C++) would enable greater parallelism and computational efficiency. This would open up the possibility of simulating multi-country systems and more detailed layers, while preserving a fine-grained micro-structure.

The remainder of this section outlines the agent taxonomy and simulation step sequence, providing a high-level conceptual map before delving into detailed module specifications in the following section.

2.1 Agent Taxonomy

To keep the model intelligible while preserving behavioral richness, all entities can conceptually be grouped into four general analytical spheres: Financial, Production, Consumption, and Policy, each capturing a distinct domain of function. Each sphere is populated by one or more agent classes that share a common decision environment and interact through markets or institutional rules. These analytical spheres are not necessarily isolated modules, but overlapping functional domains, since agents often span multiple roles, such as the Central Bank acting as both a financial actor and a policy authority. This layered design aims to reflect the institutional entanglement found in real economies.

Financial sphere The monetary backbone of the economy is formed by a single Central Bank and a population of Commercial Banks. Money enters the system endogenously through a credit creation mechanism. Reserve compliance and policy-rate transmission anchor nominal stability. Both bank types are used as accounting anchors in all monetary transactions.

Production sphere Real-sector output originates with four firm archetypes. Regular Firms convert intermediate inputs, labor, and energy into final or capital goods via recipe-defined technology. Energy Companies generate electricity using technology-specific capacity, Fossil Extractors supply primary energy carriers: coal, gas, and oil. A public Infrastructure Agency supplies non-market infrastructure services, representing the general need for roads, grids and more, financed by the government. Firms hire workers, borrow from banks, and set mark-up prices, adjust output to demand, and respond to input scarcity. Job-guarantee firms, when present, follow mission-driven investment logic and maintain open-ended hiring.

Consumption sphere Households supply labor, consume market goods and electricity, pay taxes, and store wealth in bank deposits. Income is determined by employment status and policy transfers. Preferences are modeled as utility weights over available consumption goods, drawn from a Dirichlet distribution. Consumption budgets are allocated proportionally to these weights, resulting in fixed budget shares for each good. This corresponds to a simplified linear expenditure system rather than a CES (Constant Elasticity of Substitution) utility model. While this structure does not capture substitution elasticities across goods yet, it provides a transparent and tractable first-pass basis for modeling aggregate demand and consumption flows. A CES extension remains an aim for future enhancement, although be it one requiring some deeper architectural refactoring.

Policy sphere The Government conducts fiscal policy (taxes, transfers, infrastructure spending, carbon rebates, job-guarantee wages) and issues bonds. The Central Bank, although generally located in the financial sphere, also acts as a policy agent via an inflation-targeting Taylor rule. Fiscal and monetary authorities are linked through government bond operations and balance sheet interdependence.

General Interaction logic Households supply labor to firms, receive wages, and consume goods. Firms hire labor, demand working-capital loans, and produce outputs. Banks create deposits and reserves and set interest rates. The government injects or withdraws purchasing power through taxes, transfers, spending and bond issuance.

Energy flows from Extractors to Energy Companies and ultimately to firms and households, closing the material throughput loop. All monetary, physical, and informational exchanges are logged each step, maintaining stock–flow consistency across the agent network.

2.2 Simulation Step Structure

Every discrete simulation step runs through a sequence of phases to produce an overall outcome for that step. These phases include production planning and labor allocation, electricity market coordination, financial settlements, consumption and goods exchange, policy enforcement, and balance sheet updates. This cycle reflects the dynamic feedback loops between real, financial, and institutional layers of the economy, and is designed to be both conceptually interpretable and computationally modular.

The core phases of each simulation step are as follows:

1. **Step Initialization:** Before any economic activity takes place, the simulation initializes step-specific buffers, refreshes lagged variables, and updates macroeconomic targets. Interest rates are updated using a Taylor-rule mechanism based on deviations from target inflation and output gaps. This stage ensures that agents begin each step with an internally consistent and fresh information set.
2. **Labor Market Matching:** Firms estimate labor demand based on backlog and utilization targets derived from prior production and demand conditions. Households engage in probabilistic job search using softmax routines, and firms hire applicants based on preference and capacity. The labor market embodies frictions and wage inertia, including tenure-based wage increases and wage adjustments driven by the tightness of the market.
3. **Electricity Market Operation:** All agents forecast their electricity demand for the upcoming step. A centralized day-ahead auction matches this demand with the available supply from energy companies. Prices are set based on marginal bids. This phase simulates a key input market that is sensitive to both production planning and household needs.
4. **Production and Wages:** Firms execute recipe-based production constrained by available labor, electricity, and intermediate goods. Output is determined by Leontief-type input requirements, where missing components lead to unfulfilled production. Workers receive wages at this stage, which form the primary income source for household consumption.
5. **Household Consumption and Goods Market:** Households allocate their disposable income across a consumption basket. Firms sell goods based on available inventory and current prices. This stage closes the circular flow between income generation and consumption demand, linking production to aggregate demand.
6. **Emissions Accounting:** Emissions are calculated based on fossil fuel usage in production. These feed into environmental metrics that can inform taxation or other climate policies. The emission records are tracked over time, enabling analysis of decarbonization trajectories. Emission factors are mapped from fossil energy use via fixed carbon coefficients.
7. **Financial Settlement:** Monetary dynamics are updated through interest payments, loan servicing, and defaults. Firms and households repay debt or enter bankruptcy if obligations are unmet. Commercial banks update their balance sheets, and bond transactions (coupon payments, maturities) take place. This phase formalizes the endogenous money cycle, linking credit, liquidity, and solvency.

8. **Fiscal Policy Execution:** Government engages in automatic and discretionary fiscal operations: taxing income and profits, distributing transfers. If a carbon tax is active it gets applied here.
9. **Investment and Capital Formation:** Firms decide on investment plans based on profitability, demand signals, and access to credit. Investment goods are ordered from specialized firms and used to expand productive capacity using a project pipeline logic. Public infrastructure investment is mission-driven, responding to capacity targets and systemic needs.
10. **Market Entry and Exit:** The economy is open to structural transformation via endogenous firm turnover. New firms may enter based on signals such as unmet demand or rising markups, while unprofitable firms may exit.
11. **Post-Production Updates:** Depreciation reduces capital stocks with a grace period for new investments. Firms update prices based on input costs and shortages using markup rules. Wage adjustments reflect persistent labor market trends, anchoring nominal dynamics to real economy frictions.
12. **Regulatory Compliance:** Banks are checked for reserve compliance, ensuring monetary stability. Unsold electricity is discarded, reflecting physical constraints. This phase embeds institutional discipline into the model, enforcing rules that constrain agent behavior.
13. **Macro Metric Recording:** Key indicators — GDP, inflation, employment, income distribution, energy mix — are calculated and logged. These metrics enable dynamic analysis of economic performance and policy effects, facilitating validation and calibration.
14. **Sectoral Balance Snapshot:** Stock-flow consistency is verified by recording balance sheets across all institutional sectors (households, firms, banks, government, central bank) and aggregating them into Godley-style sectoral balances. These snapshots provide insight into monetary dynamics, credit expansion, and sectoral interdependence.
15. **Visualization and Monitoring:** If enabled, the system updates visualization panels displaying time series of core indicators. This enhances transparency and aids in model debugging or exploratory analysis.
16. **End-of-Step Cleanup:** Signals are updated for the next round, including fuel availability and entry signals. Temporary tracking fields are cleared, and the simulation prepares for the next monthly cycle.

Each of these stages is handled by a dedicated high-level function, ensuring that the model remains transparent, testable, and extensible. Execution order is currently essential, as inter-phase dependencies exist. For instance, production cannot proceed without prior allocation of electricity, hence the electricity market phase needs to precede production. This is a result of the currently sequential simulation architecture, which could in future iterations be restructured toward finer-grained or parallelized execution, enabling more realistic asynchronous interactions and the study of real-time coordination dynamics.

3 Model Specification

This section translates the conceptual architecture introduced above into its operational components. Each subsection describes a functional subsystem of the model, detailing the decision rules, data structures, and parameter settings that govern agent behavior and macroeconomic dynamics. Emphasis is placed on how key features like stock-flow consistency, input-output dependencies,

and market coordination are implemented. Where applicable, relevant algorithms and parameter values are included to document the model’s logic with transparency and reproducibility. This formal specification also serves as a reference point for later discussions on scenario design and result interpretation.

3.1 Goods System and Recipe-Based Production

The real economy backbone is organized around a hierarchical goods catalog and a recipe-driven production engine. Together these elements create supply-chain dependencies, resource constraints, and technology switching without forcing the user to specify explicit input–output matrices *ex ante*. Large-scale ABMs such as Poledna et al. [2022] embed detailed ESA input–output tables to reproduce sectoral linkages. In this thesis IO tables are instead built from a stochastically generated goods catalog and a tier-respecting recipe generator. Unlike fixed national IO tables, the stochastic catalog creates an ensemble of plausible supply networks. Running each policy scenario on many such worlds reveals general trajectories and paths that can show whether a result is structural or merely an artefact of one particular set of technical coefficients. This approach also scales effortlessly: adding a new good or production technology only requires an additional draw from the generator, while new or policy-relevant goods can be added manually without re-balancing the entire IO structure.

Both layers, goods and recipes, are created once at model initialization and then remain fixed for the entire simulation horizon. Coefficients are calibrated *ex-post* to satisfy a Leontief feasibility threshold (τ) [Leontief, 1986] while preserving random draw proportions. The stochastic process determining recipes and, by extension, production technology follows a three step process:

1. Catalog generation A `generate_good_types( $n$ )` function inside the model draws n goods, assigning each an initial utility weight. Specialized generators add one investment good for capacity expansions, maintenance goods, three fossil fuels, and two intermediate energy carriers (Electricity, Hydrogen). All entries are stored as `GoodType` records, providing a single source of truth for every downstream module.

2. Tier assignment (Acyclic Supply Graph) A deterministic map `tier_map : good_id  $\rightarrow$  {0, 1, 2}` turns the flat list into a directed-acyclic structure [Miller and Blair, 2009]:

- Tier 0 — *resources*: fossil fuels and stylized raw materials.
- Tier 1 — *intermediates*: maintenance goods and randomly chosen intermediate goods.
- Tier 2 — *finals*: remaining consumables and the generic Investment Good type.

Energy carriers (Electricity, Hydrogen) are flagged tier = -1 and may appear as inputs at any level. The tier map guarantees that material flows never form direct cycles, so recipe generation can be automated without manual IO tables.

3. Recipe generator For every non-energy good the recipe generator creates one to three fixed-coefficient recipes with the following rules in place:

- a) Electricity is mandatory. Consumption is derived from machine-hour estimates and a kWh/machine–h draw.
- b) With probability 0.4 an additional fuel (coal, oil, gas) or hydrogen input is added. All recipes sharing such an energy input belong to a common `substitution_group`, enabling later fuel-switching at run time.
- c) Remaining material inputs are sampled from lower-tier goods only, preserving the structure of a direct acyclic graph.

- d) Labor (L) and machinery (M) coefficients are drawn from tier-specific ranges and then normalized so that

$$\sum \text{inputs} + L + M \leq \tau,$$

where $\tau \approx 3$ units defines one “budget” of physical effort.

Because coefficients are random but subsequently scaled, every recipe is feasible by construction. Later calibration of τ or the labor multiplier rescales all recipes coherently.

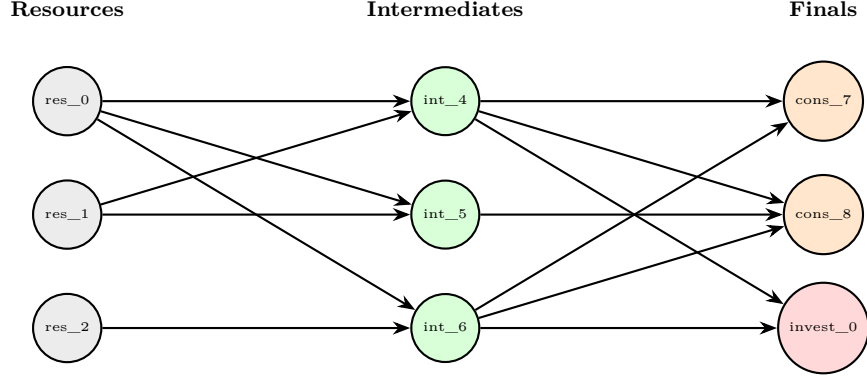


Figure 1: Toy example of the tiered recipe graph with randomly generated goods. Each intermediate takes multiple resource inputs, finals combine different intermediates. Excluding general fuel and electricity inputs for visual clarity.

To capture electricity requirements across diverse production processes, each recipe includes two components of power demand. First, a “process heat and auxiliary” draw is sampled uniformly between 0.05 and 0.2 kWh per unit of output, representing non-productive electricity used for lighting, controls, and auxiliary systems [IEA, 2014]. Second, machine-hour consumption is modeled at 3–7 kWh per machine-hour to reflect industrial machinery use [IEA, 2014]. Both ranges are to be interpreted as simple and generally conservative starting heuristics for a small-scale test system with only 200 households, limited production realism, and a constrained set of goods. I assign coal 24 MJ/kg, natural gas 54 MJ/kg, and oil 42 MJ/kg as lower heating values, consistent with reported baselines [Krey et al., 2014, International Energy Agency and Organisation for Economic Co-operation and Development, 2005]. Fossil fuel conversion into electricity assumes thermal efficiencies of 38% for coal, 42% for oil, and 50% for gas-fired power plants, in line with typical dispatchable generation technologies [IEA, 2019]. Hydrogen recipes use an electrolysis energy intensity of roughly 50–60 kWh/kg, consistent with published ranges for PEM and alkaline systems [Aghakhani et al., 2023]. The CO₂ emission factors for fossil fuels are set to 0.00242 tCO₂/kg for coal, 0.00275 tCO₂/kg for natural gas, and 0.00315 tCO₂/kg for oil, following IPCC AR5 Annex II recommendations [Krey et al., 2014]. All physical unit conversions (e.g., 1 kWh = 3.6 MJ) follow SI conventions. Finally, all recipes are normalized by a feasibility threshold $\tau = 3.0$, ensuring the sum of physical inputs (materials, electricity, labor, machinery, maintenance) remains within plausible bounds without distorting relative proportions.

3.2 Production System

The production system operationalizes the recipes introduced in the previous section by simulating firms that hire labor, procure inputs, and convert them into outputs through constrained processes. This section outlines the full production cycle from hiring to inventory addition, including the associated constraints (labor, energy, capital) and integration with markets.

3.2.1 Labor Market Dynamics

The labor market is executed in every simulation step immediately before the production phase. Its purpose is to (i) translate firm-level production targets into workforce demand, (ii) probabilistically match unemployed households to vacancies, (iii) adjust contract wages in response to profitability and hiring frictions, and (iv) update tenure and employment status variables used elsewhere in the model (e.g. for household income, skill accumulation, or wage bargaining).

The high-level sequence is

- L1) Reset employment flags** of all households;
- L2) Labor-demand estimation** for each firm f ;
- L3) Soft-max job matching** between vacancies and unemployed workers;
- L4) Hiring** (vacancy fulfillment) and status updates;
- L5) Wage adjustments** driven by profits and labour shortages;
- L6) Cleanup** of temporary data structures.

The following subsections formalize each block.

Workforce Demand Estimation The first task of the labor-market phase is to translate output plans into a headcount request. A firm f selects a production target Q_f^{tar} (physical units per step) and, given the fixed labor intensity of its active recipe, derives the associated hours and headcount. Let ℓ_f be the labor coefficient of the recipe in h/unit and H_{step} the number of working hours in one simulation step¹, then the raw workforce requirement is:

$$L_f^{\text{raw}} = \left\lceil \frac{Q_f^{\text{tar}} \ell_f}{H_{\text{step}}} \right\rceil \quad (1)$$

Firms cannot request fewer than a technical minimum crew L_f^{min} and may be capacity-constrained by L_f^{max} to avoid infinite scaling. The final demand is therefore:

$$L_f^{\text{desired}} = \min \left\{ \max(L_f^{\text{raw}}, L_f^{\text{min}}), L_f^{\text{max}} \right\} \quad (2)$$

The definition of Q_f^{tar} depends on the information available in each sector. Four stylised cases are distinguished:

a) Standard consumer-goods firms

Decision making is anchored on three observable signals. First, the firm forms an expectation of next-period sales via an exponentially weighted moving average of realized turnover [Hynman and Athanasopoulos, 2021], denoted $s_f^{\text{smooth}} = \beta q_{t-1} + (1 - \beta)s_{f,t-1}$ with smoothing factor $\beta \in (0, 1)$ (baseline $\beta = 0.3$). Second, it monitors its inventory position relative to a target buffer $I^* = \kappa s_f^{\text{smooth}}$. The coverage ratio κ is set to 0.5, corresponding to “half a step” of protective stock. Third, it records the backlog B_f of unfilled customer orders that were forwarded by the goods-market clearing routine.

A relative price term adjusts the perceived sales signal: if the firm’s own price $p_{f,t}$ deviates from the sectoral median $\tilde{p}_{f,t}$, demand is discounted or boosted by a factor $\delta_f^{\text{price}} = p_{f,t}/\tilde{p}_{f,t} - 1$. To avoid blow-ups when prices are temporarily far apart, the deviation is bounded to the

¹In the baseline calibration one step represents one month, so $H_{\text{step}} = 160$ h at 40h/week. Changing this parameter implicitly changes the time a step represents.

interval $[-0.5, 0.5]$. This ensures that even a large relative price gap only adjusts the sales signal by at most $\pm 50\%$, preventing extreme swings in production targets due to transient price spikes or drops.

Combining the three drivers gives the production target

$$Q_f^{\text{tar}} = \underbrace{\alpha s_f^{\text{smooth}} (1 + \delta_f^{\text{price}})}_{\text{expected sales}} + \underbrace{\lambda (I^* - I_{f,t-1})}_{\text{inventory gap}} + \underbrace{\phi B_f}_{\text{backlog}} \quad (3)$$

with three behavioral parameters: α (*sales sensitivity*, baseline 1.1) lets firms overshoot expected demand, λ (*inventory correction rate*, 0.1–0.3) controls how aggressively the stock gap is closed, ϕ (*backlog urgency*, 0–1) weights the priority given to clearing queued orders. Substituting Q_f^{tar} into Eqs. (1)–(2) produces the final head-count request L_f^{desired} for the matching stage.

b) Investment–goods firms

Because capital equipment is ordered in large, irregular batches, step based sales data contain little useful information. The sector therefore bases its output target on two slowly moving stock variables:

- (i) the *economy-wide order book* D^{inv} , implemented in the code as the global parameter `unmet_investment_demand`. It increases whenever downstream firms attempt to buy kits that are unavailable and declines as kits are delivered;
- (ii) the firm-specific *installation backlog* B_f^{inv} , equal to the number of kits that have been sold but are still somewhere in the `investment_pipeline` awaiting delivery or installation.

With these definitions the production target is

$$Q_f^{\text{tar}} = \underbrace{\sigma D^{\text{inv}}}_{\text{sector backlog}} + \underbrace{\chi B_f^{\text{inv}}}_{\text{own backlog}}, \quad (4)$$

where σ (*pipeline responsiveness*) translates the global shortfall into new output and χ (*clear-your-own-queue rate*) converts private backlog into planned production. Typical calibrations use $\sigma \approx 1.0$ and $\chi \approx 0.10$, meaning the sector aims to remove the system backlog in one step while each firm clears about ten per cent of its own queue per step.

After computing Q_f^{tar} the head-count request L_f^{desired} is again obtained via Eqs. (1)–(2).

c) Infrastructure service providers

Infrastructure service is a for now simplified good supposed to capture generic, economy-wide support functions, transport and logistics infrastructure, that every production recipe requires in proportion to its real activity. Each recipe carries a fixed coefficient

$$\iota_j = \text{SI} \bar{\iota}, \quad \bar{\iota} \approx 0.01,$$

where SI is a global service-intensity tuning knob that will be elaborated on in the Model Calibration section. At the beginning of the labor phase the model loops over all producing firms, multiplies last-period output q_j^{prod} by ι_j , and aggregates the result:

$$D_t^{\text{infra}} = \sum_{j \in \mathcal{F}} q_j^{\text{prod}} \iota_j, \quad \mathcal{F}: \text{set of all production firms.}$$

The value D_t^{infra} therefore rises when the real side of the economy expands and falls during recessions.

If N_{infra} identical infrastructure providers exist, each firm receives an equal share

$$Q_f^{\text{tar}} = \frac{D_t^{\text{infra}}}{N_{\text{infra}}}$$

Eq. (1) then converts Q_f^{tar} into a head-count request, ensuring that total service capacity scales one-for-one with real demand.

In the first steps of a simulation, when no historical production data exist, D_t^{infra} is initialized from a scalar estimate of infrastructure spending embedded in the experiment parameters. This avoids an artificial bottleneck at start-up.

A separate routine compares realized service supply $\tilde{D}_t^{\text{infra}}$ with demand D_t^{infra} . Persistent shortfalls reduce an economy-wide efficiency multiplier $\theta_t \in (0, 1]$ that scales the effective capital input of all recipes, while persistent surpluses gradually raise θ_t toward 1. The hiring rule above therefore couples labor demand in the infrastructure sector to the same efficiency feedback that it ultimately helps to determine.

d) **Electricity generators**

Generation firms do not rely on their own sales history; instead they receive a system demand forecast D_t^{elec} from the electricity-market phase.² If N_{gen} generators participate in the same balancing area, a first-round allocation gives each the symmetric target:

$$Q_f^{\text{tar}} = \frac{D_t^{\text{elec}}}{N_{\text{gen}}}$$

Because recipe coefficients are defined per unit of name-plate capacity, a stylized photovoltaic park has a much smaller raw labor coefficient ℓ_f than e.g. a gas turbine. Solar panels need only routine inspection, while thermal plants require more continuous staffing. To express labor input on a per-kWh delivered basis the technology’s capacity factor is divided by $c_f \in (0, 1)$:

$$\ell_f^{\text{eff}} = \frac{\ell_f}{c_f}$$

A low-utilization technology keeps its staffing advantage even after the adjustment.

If a plant’s existing capacity is smaller than Q_f^{tar}/c_f , the gap triggers the investment routine described in Section 3.4. Conversely, if capacity is plentiful, Q_f^{tar} may be met with fewer labor hours, mimicking dispatch choices that idle the least-efficient units first.

For every sector the production target Q_f^{tar} feeds into the labor-hours formula (1) and the crew constraint (2), yielding the desired head-count L_f^{desired} forwarded to the matching algorithm. The numerical weights $(\alpha, \lambda, \phi, \sigma, \chi, \dots)$ are, for now, filled with placeholder values used in small-scale test runs. In larger-scale experiments the model will draw these parameters at random from calibrated ranges on a per firm level to capture heterogeneous behavior.

Probabilistic Job Matching The model currently treats job search as a one-shot discrete-choice problem [Train, 2002] that occurs after firms have posted their vacancies and before wages

²The forecast is currently the total kWh requested by all non-energy firms after their production planning plus a fixed “baseline” floor that keeps the grid energized when real-side demand is very low.

are revised. Let the vacancy set be

$$\mathcal{V} = \{(f, v_f, w_f)\}, \quad v_f = L_f^{\text{desired}} - L_{f,t-1} > 0,$$

where v_f is the number of open positions at firm f and $w_f = \ln(\text{wage}_f)$ the log of the posted hourly wage.³ Every unemployed household h draws exactly one application destination. The probability of picking firm f is a soft-max (logit) function of the posted wage [McFadden, 1974],

$$P_{h \rightarrow f} = \frac{\exp[(w_f - w_{\max})/(T w_{\max})]}{\sum_{g \in \mathcal{V}} \exp[(w_g - w_{\max})/(T w_{\max})]}, \quad w_{\max} \equiv \max_{g \in \mathcal{V}} w_g, \quad (5)$$

with temperature parameter $T > 0$:

- $T \rightarrow 0$ – nearly all workers flock to the top-paying firm (greedy search);
- $T \rightarrow \infty$ – applications are spread almost uniformly (random search).

The normalization by $w_{\max}$ keeps the argument of the exponential in the interval $[-1/T, 0]$, preventing numerical overflow even when wage dispersion is large.

Algorithmic flow:

- i) Build $\mathcal{V}$ once, storing current $\{w_f\}$ and residual vacancies $\{v_f\}$.
- ii) Iterate over households that are not marked **employed_this_step**. Draw a destination according to (5).
- iii) Add the household to the chosen firm’s applicant pool and decrement v_f . If v_f reaches 0, remove $(f, \cdot, \cdot)$ from $\mathcal{V}$ so later workers cannot apply to a filled firm.

The matching routine touches every unemployed household exactly once: for each person it looks up the pre-calculated wage scores of all firms and draws a random application. If the simulated population doubles, the algorithm therefore takes about twice as long. An optimization task for a future model iteration that needs to rework labor-dynamics in more depth.

Equation (5) can be viewed as a reduced-form model of job search with incomplete information and idiosyncratic preference shocks. The temperature T calibrates how strongly wages dominate other (unmodeled) considerations such as commuting distance, firm reputation, or others. A moderate baseline of $T = 0.2$ is used in test runs.

Once the application phase is finished, firms select from their applicant pools up to the residual gap v_f . Households that remain unmatched stay unemployed until the next step or enter policy scenarios with job-guarantee employment (introduced later).

Hiring, Employment Update, and Wage Formation After applications are collected, each firm iterates through its applicant list and hires up to the residual gap

$$m_f = \min\{v_f, |\text{applicants}_f|\},$$

³A practical reason motivates the log transformation. Raw wages in the soft-max exponent can explode. For example, with $T = 0.01$ and wages of €40 versus €15,

$$\frac{40 - 15}{0.01 \cdot 40} = 62.5, \quad \exp(62.5) \approx 7.5 \times 10^{27},$$

which far exceeds floating-point limits. By instead using log-wages ($\ln 40 - \ln 15 \approx 0.98$) and normalizing by the max log-wage, exponent arguments shrink dramatically—e.g. with $T = 0.2$ they lie in $[-5, 0]$, so exp ranges only from $\exp(-5) \approx 0.007$ to 1, safely within machine precision.

where $v_f = L_f^{\text{desired}} - L_{f,t-1}$ is the number of still-open positions. Applicants are processed in arrival order⁴ and flagged as employed for this step.

If a firm cannot fill every vacancy ($m_f < v_f$) the shortfall carries forward into the next planning round and will trigger a wage response. For every hired worker h the model sets

$$\text{employer}_h \leftarrow f, \quad \text{tenure}_h \leftarrow \text{tenure}_h + 1,$$

and clears the transient field `applied_to`. Households that remain unmatched keep `employer = None` and `tenure = 0` for this step.

Monthly pay is revised through *three* overlapping mechanisms:

(i) Profitability lift. If last-period revenue exceeded the sum of costs ($R_f > C_f$) the firm raises wages by a factor $(1 + \eta)$, with $\eta > 0$ calibrated between 1% and 3%.

$$w_{f,t+1} = w_{f,t}(1 + \eta) \quad \text{if } R_f > C_f.$$

(ii) Shortage premium. If the firm could not fill its vacancies ($L_{f,t} < L_f^{\text{desired}}$) and was not profitable, it nevertheless grants a smaller raise—half the standard step size η —to make the position more attractive:

$$w_{f,t+1} = w_{f,t}\left(1 + \frac{1}{2}\eta\right) \quad \text{if } R_f \leq C_f \wedge L_{f,t} < L_f^{\text{desired}}.$$

(iii) Tenure ladder. For each employee h the contract must obey

$$w_{f,t+1} \geq w_{\min}\left(1 + \rho \lfloor \tau_h/12 \rfloor\right),$$

with $w_{\min}$ the statutory minimum wage, τ_h the worker’s tenure in months, and ρ an annual step increase ($\approx 1\%$).

All updated wages are clipped below by $w_{\min}$ and feed back into the next soft-max matching round. Unfilled vacancies raise labor demand in the subsequent step, creating a feedback loop among hiring, production, and pay. Overall these dynamics would have to be reworked in a future iteration however. Job search, wage dynamics and tenure are still oversimplified mechanisms in the current version. The model currently also lacks a concept of skills and qualifications: all labor is treated as equal in nature and providable by every worker. Transitioning from this state to more realistic behavior would however again warrant a bigger refactor.

3.2.2 Production Phase

Once vacancies have been filled the simulation advances to production. At this point each firm f knows

- (i) its workforce $L_{f,t}$ delivered by the labour market and
- (ii) if it is an electricity generator: the dispatch schedule produced during the electricity-market phase (Section 3.2.3).

These inputs are transformed into material output, inventories, and payments as summarized in Figure 2. The present sub-section details the generic routine shared by all firms except generators and infrastructure providers, whose sector-specific adaptations follow after.

⁴Arrival order is a deterministic proxy for ‘first come, first served’ and avoids an additional random draw.

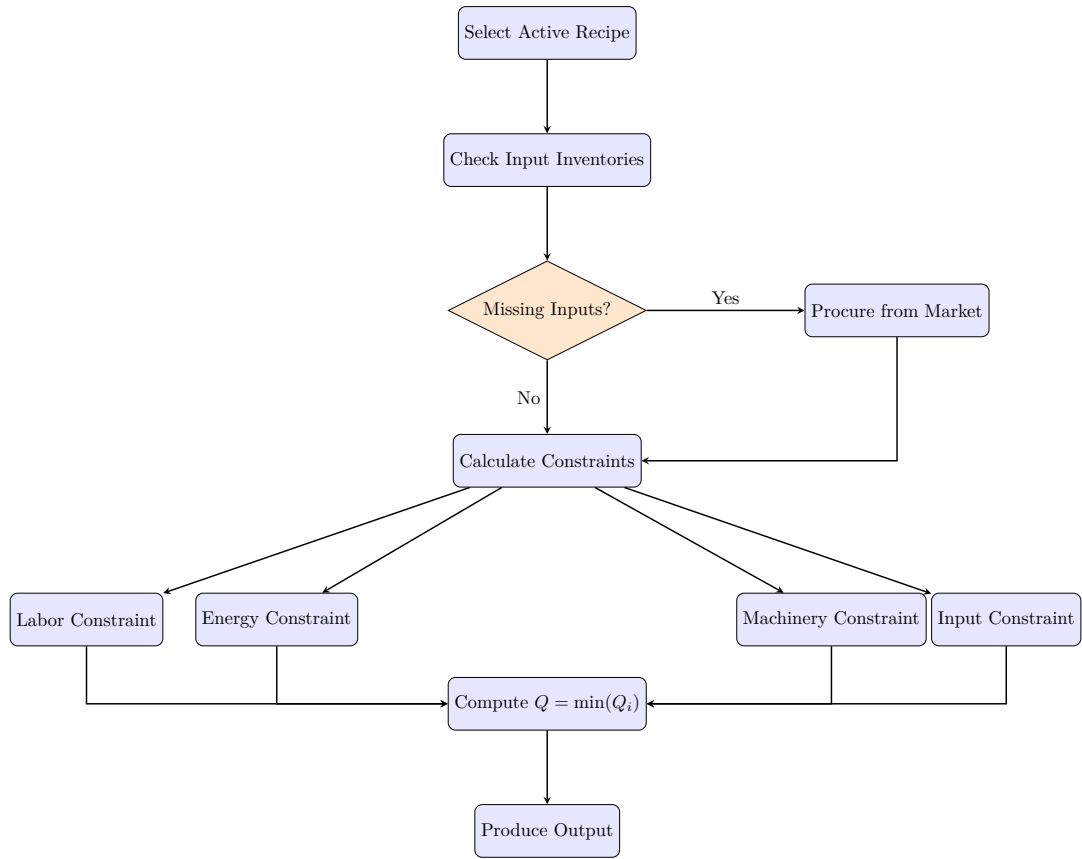


Figure 2: Firm-level production flow. After selecting a recipe and checking inventories, firms compute constraints and determine feasible output. Output is limited by the most restrictive of labor, energy, machinery, or input availability.

High-level sequence

- P1) Reset diagnostics** Clear all per-step counters (e.g. output, input spending).
- P2) Apply infrastructure efficiency** Read the current economy-wide efficiency ratio θ_t and scale the recipe's labor and input coefficients for this step.
- P3) Select active recipe** Pick the cost-minimizing process $\hat{\rho}_f$.
- P4) Procure missing inputs** Purchase any shortfalls of ingredients required by $\hat{\rho}_f$, borrow if deposits are insufficient.
- P5) Evaluate capacity ceilings** Compute labor, machinery, energy, and materials constraints and set feasible output to the most restrictive limit.
- P6) Execute & bookkeep** Consume inputs, create output goods, pay wages, and write statistics for downstream modules.

Infrastructure-adjusted coefficients The model treats the state of public infrastructure as a neutral productivity shifter, entering the aggregate production function as a scalar that modifies output without changing marginal rates of substitution [Solow, 1957]. The infrastructure block contains

- (i) an *efficiency ratio* $\theta_t \in [\theta^{\min}, 1]$ that compares services supplied by the infrastructure agency with services demanded as a result of production by all other firm type agents in the previous step, and

(ii) an *average quality index* $q_t \in [0, 1]$ of the physical network.

Since infrastructure services are injected into every production recipe and therefore required as an input, the amount of infrastructure needed for efficient production scales with overall system output. At the start of every production step the infrastructure module computes an economy-wide service ratio. A value $\theta_t < 1$ is interpreted as congestion, detours, or delays that erode factor productivity based on missing capacity. The scheduler converts this macro signal into firm-level 'effective' coefficients that are applied on a per step basis:

$$\text{Labor: } \ell_f^{\text{eff}} = \frac{\ell_f}{\theta_t}, \quad (6)$$

$$\text{Materials: } a_{g,f}^{\text{eff}} = \begin{cases} a_{g,f}, & \theta_t \geq \theta_{\text{crit}}, \\ a_{g,f} [1 + \xi (1 - \theta_t)], & \theta_t < \theta_{\text{crit}}, \end{cases} \quad (7)$$

ℓ_f Engineering labor hours per unit in the recipe file.

$a_{g,f}$ Baseline quantity of input g required per unit.

θ_{crit} Congestion threshold that triggers extra input demand (default 0.8).

$\xi \in [0, 0.5]$ Slope that governs how quickly extra materials rise as θ_t falls (default 0.5).

The quality of the infrastructure network does not modify production coefficients directly, instead it modulates the amount of service hours that count toward the aggregate supply side of the ratio θ_t :

$$\theta_t = \frac{\sum_{f \in \mathcal{I}} q_{f,t} D_{f,t}}{\sum_{f \in \mathcal{I}} D_{f,t}}, \quad \mathcal{I} : \text{set of infrastructure providers}, \quad (8)$$

where $D_{f,t}$ are the infrastructure services actually produced this step. Quality depreciates geometrically and improves through maintenance inputs purchased by the infrastructure firms:

$$q_{t+1} = \min\{1, q_t (1 - \delta) + \rho_L M_t^{\text{light}} + \rho_H M_t^{\text{heavy}}\},$$

with random wear-rate $\delta \sim \mathcal{U}(0.005, 0.01)$ and repair efficiencies $\rho_L < \rho_H$.

This design treats infrastructure as a common-pool resource that feeds back into every private production while adding only a simple multiplicative operation to the inner loop. Since the infrastructure agency in the model needs to actively maintain their service and is funded by the government agent, the mechanism also creates a needed channel for continuous public-sector spending. This 'one-size-fits-all' representation should eventually be replaced by a more granular treatment of infrastructure needs however.

Recipe choice Every output in the model can be produced by several alternative recipes. At the start of every step each firm selects one active recipe $\hat{\rho}_f$ to be used throughout the subsequent procurement and production blocks. The decision is purely cost-driven but tempered by two behavioral features: noisy information about input prices and a reluctance to switch technologies too often. The sequence follows these steps:

R1) Enumerate candidates: collect all recipes that deliver the firm's output good.

R2) Update expected marginal costs: For any recipe r operated by firm f the model treats marginal cost as the current-price value of all variable factors that are consumed

to produce one physical unit.⁵ With $p_{g,t}$ the prevailing market price of input g in step t , $w_{f,t}$ the firm's hourly wage, and p_t^{el} the clearing price of electricity, the formula is

$$c_{r,t} = \underbrace{\sum_{g \in \mathcal{G}_r} a_{g,r} p_{g,t}}_{\text{materials}} + \underbrace{\ell_r w_{f,t}}_{\text{labour}} + \underbrace{e_r p_t^{\text{el}}}_{\text{electricity}} + \underbrace{\mu_r^L p_t^{\text{maint,L}} + \mu_r^H p_t^{\text{maint,H}}}_{\text{light/heavy maintenance}} + \underbrace{\kappa_r \pi_t^{\text{CO}_2}}_{\text{carbon charge}}, \quad (9)$$

where

$a_{g,r}$ quantity of material input g per unit,

ℓ_r labour hours per unit,

e_r kWh of electricity per unit,

μ_r^L, μ_r^H light / heavy maintenance goods per unit,

κ_r kg CO₂ emitted per unit,

$\pi_t^{\text{CO}_2}$ carbon price applied that step.

The resulting $c_{r,t}$ is stored, smoothed, and used in the next step's recipe-choice decision with:

$$\tilde{c}_{r,t} = (1 - \gamma) \tilde{c}_{r,t-1} + \gamma c_{r,t}, \quad 0 < \gamma \leq 1,$$

and $\gamma \approx 0.3$ in the baseline calibration. Smoothing is applied here so that eventual price spikes do not lead to rapid recipe switching behavior. As this step is done before the actual procurement takes place this also serves as an adaptive forecast.

R3) Feasibility filter: After ranking recipes by expected cost the firm keeps only those that can be executed immediately. Two deterministic screens are applied:

- (a) *Labor screen.* The hours available in the current step, $L_{f,t} H_{\text{step}}$, must cover the recipe's requirement ℓ_r :

$$L_{f,t} H_{\text{step}} \geq \ell_r.$$

If this inequality fails the recipe is infeasible until additional workers are hired.

- (b) *Inventory screen.* For every critical input (all non-energy goods) the on-hand stock must exceed a 10 % reserve:

$$S_{g,f}^{\text{inv}} \geq 0.1 a_{g,r}, \quad g \notin \{\text{Electricity, Hydrogen, Coal, Gas, Oil}\}.$$

Recipes that would deplete a critical stock below the safety margin are postponed until the next procurement round.

The surviving set is passed to the switch rule, which decides whether the cheapest feasible candidate warrants the organizational cost of a process change.

R4) Cost ranking Orders the feasible candidates by their expected marginal costs $\tilde{c}_{r,t}$ and identify the cheapest one. If the returned list is empty the firm has no executable recipe this step and remains idle.

⁵Fixed capital costs are handled separately through depreciation or maintenance and therefore do not enter the marginal cost.

R5) Hysteresis test : Given $\tilde{c}_{\text{old},t}$, the expected marginal cost of the recipe that was active in $t - 1$, switching incurs organizational costs (re-tooling, training, certification) that the model summarizes in a single switch threshold $\eta_s > 0$. A change is implemented only if the relative gain exceeds this hurdle:

$$\frac{\tilde{c}_{\text{old},t} - \tilde{c}_{r^*,t}}{\tilde{c}_{\text{old},t}} > \eta_s, \quad \eta_s \simeq 0.05 \text{ (baseline)}.$$

With $\eta_s = 0.05$ the firm demands at least a 5% cost advantage before abandoning its current technology, preventing ping-pong behavior when marginal costs are very similar.

If the inequality holds, the firm adopts r^* and registers the change as a recipe switch for the analytics module, otherwise it keeps the current process for the step. This procedure generally aligns with an evolutionary economics approach [Nelson and Winter, 1982], treating technological change as path-dependent and incremental. Currently all recipes are visible for all companies at the same time however. Extending recipe logic with an R&D based discovery logic could make for a valuable addition in terms of realism.

Input procurement Given the intended output plan Q_f^{plan} and the infrastructure-adjusted coefficients $a_{g,f}^{\text{eff}}$, the gross requirement of every input g equals:

$$\hat{S}_{g,f} = a_{g,f}^{\text{eff}} Q_f^{\text{plan}}$$

Comparing this value with the firm's on-hand inventory $S_{g,f}^{\text{inv}}$ yields the shortage:

$$\Delta S_{g,f} = \max\{0, \hat{S}_{g,f} - S_{g,f}^{\text{inv}}\}$$

Whenever $\Delta S_{g,f} > 0$ a helper routine `_procure()` is called. Its logic can be summarized in four ordered steps:

- (i) **Inventory sweep :** Search the market's list of finished-goods lots (a pool shared by all firms), starting with the lowest asking price, until $\Delta S_{g,f}$ is covered or no lots remain.⁶
- (ii) **Standing contracts :** If the good is a primary (tier-0) input and inventories are exhausted, place a back-to-back order directly with one of its producers at the current list price.
- (iii) **Payment and credit :** Each purchase draws on the firm's deposits. When liquidity is insufficient the firm invokes the borrowing helper `borrow_if_needed()`, which attempts to arrange an amortizing loan with its commercial bank.
- (iv) **Residual gap :** Any quantity that remains uncovered after steps (i)–(iii) becomes an *input constraint*; it enters the capacity-ceiling calculation (step P5).

The procedure is repeated for every input of the active recipe $\hat{\rho}_f$ before production proceeds.

Capacity limits After inputs have been secured, four independent ceilings bound the amount a firm can physically deliver in the current step. Using the effective coefficients and H_{step} working

⁶A depth-first search avoids sorting the entire pool and is therefore $\mathcal{O}(\Delta S)$ in the number of units bought.

hours per step, the limits read

$$\begin{aligned}
\text{labour: } Q_f^{\text{lab}} &= \frac{L_{f,t} H_{\text{step}}}{\ell_f^{\text{eff}}}, \\
\text{machinery: } Q_f^{\text{mach}} &= \frac{K_f H_{\text{step}}}{\kappa_f}, \\
\text{electricity: } Q_f^{\text{elec}} &= \frac{E_f^{\text{inv}}}{e_f^{\text{eff}}}, \\
\text{materials: } Q_f^{\text{inp}} &= \min_g \frac{S_{g,f}^{\text{inv}}}{a_{g,f}^{\text{eff}}}.
\end{aligned}$$

The firm cannot exceed the most restrictive bound, hence output quantity becomes:

$$Q_{f,t} = \min\{Q_f^{\text{lab}}, Q_f^{\text{mach}}, Q_f^{\text{elec}}, Q_f^{\text{inp}}\}.$$

Taken together, the four ceilings Q_f^{lab} , Q_f^{mach} , Q_f^{elec} , and Q_f^{inp} form a hierarchy of hard constraints. Output can expand with additional labor only if spare machine- hours, kWh and inventories are simultaneously available. Conversely, a shortfall in any single factor reduces realized production one-for-one. There is no substitution between factor inputs, imbalances are not smoothed internally but propagate immediately to the macro level through a lower $Q_{f,t}$.

Goods produced in quantity $Q_{f,t}$ are instantiated as `real_assets`, input stocks are debited, and wages are paid. The model logs marginal costs, carbon emissions, and value flows for the national-accounts module.

Electricity generators follow the same general accounting loop as ordinary firms except that their physical output is already produced in the electricity-market phase. Which is a structural requirement in the sequential setup, since the other firms can not produce without having their energy needs covered first.

The described logic of hard production ceilings deliberately avoids factor substitution, emphasizing fixed input-output proportions and direct propagation of input shortfalls to the macroeconomic level. This modeling choice is made to highlight and extend on bottlenecks and resource constraints in more depth in future iterations.

3.2.3 Electricity-market phase

The power market is executed after the labor phase and before ordinary production. Its task is to (i) forecast economy-wide electricity demand for the next step, (ii) schedule generation so that the grid is energized when non-energy firms enter their production loop, and (iii) discover a uniform day-ahead price via a wholesale auction.

High-level sequence:

- E1) Demand forecasting** Households form a price-elastic request, firms compute process-based kWh bids.
- E2) Capacity planning** Generators receive the aggregate forecast and pre-commit to a proportional share, adding a +2% safety margin but never exceeding name-plate capacity.
- E3) Merit-order dispatch** Each generator internally allocates its share across installed technologies based on marginal cost.

- E4) Uniform-price auction** All buyers (households and firms) and all generators participate in a single clearing. Allocations are pro-rata when demand exceeds supply; the clearing price is paid to every accepted kWh.
- E5) Settlement & feedback** Deposits move between agents and banks, generators' list prices are overwritten by the clearing price, shortage metrics update the adaptive safety-margin rule.

Demand formation: In each time-step two blocks of agents enter the day-ahead market. Households form a quantity bid that depends on a baseline requirement, idiosyncratic taste shocks, and the previous market price,

$$d_{h,t} = \bar{d}(1 + \varepsilon_h)(p_{t-1}^*)^{\varepsilon_p}, \quad \varepsilon_h \sim \mathcal{U}[-0.15, 0.15], \quad \varepsilon_p = -0.4.$$

The baseline $\bar{d}$ is fixed at 150 kWh per month, around the EU average for one household [Odyssee-MURE, 2024]. A negative price-elasticity ε_p makes demand fall when electricity was expensive in the previous clearing; by keying on p_{t-1}^* rather than the not-yet-known current price the sequential structure of the model is preserved.

Non-energy firms derive their bids mechanically from the production plans determined in the labor phase. If $\hat{e}_f$ is the electricity coefficient of the recipe chosen by firm f and $Q_{f,t}^{\text{plan}}$ its intended physical output then the forward purchase is

$$b_{f,t}^{\text{elec}} = \hat{e}_f Q_{f,t}^{\text{plan}}.$$

Because $Q_{f,t}^{\text{plan}}$ implicitly already incorporates input inventories and labor ceilings, firms never over-buy deliberately; any shortfall observed later in the production phase therefore reflects a system-wide auction shortage rather than miscalculation by individual firms.

Total market demand is simply the sum of household and firm bids, $D_t^{\text{elec}} = \sum_h d_{h,t} + \sum_f b_{f,t}^{\text{elec}}$, and is passed to the generator block for capacity planning.

Generation commitment: The commitment routine bridges the demand forecast from the previous subsection and the plant-level dispatch executed inside each **EnergyCompany**. Let K_f denote generator f 's name-plate capacity in MW and H_{step} the duration of one simulation step in hours (baseline $H_{\text{step}} = 160$ for a “working month”). Expressed in energy units, system-wide technical potential is

$$\bar{K}_t = \sum_{g \in \mathcal{G}} K_g \cdot 1000 H_{\text{step}} \quad [\text{kWh}],$$

with $\mathcal{G}$ the current set of generators and the factor 1000 converting MW to kW.

Given total demand D_t^{elec} the model, via an implicit Independent System Operator (ISO), plans to cover it plus a thin 2% safety margin, but never schedules more than the physical ceiling:

$$Q_t^{\text{plan}} = \min\{1.02 D_t^{\text{elec}}, \bar{K}_t\}.$$

The target is then distributed to installed capacity,

$$Q_{f,t}^{\text{plan}} = Q_t^{\text{plan}} \frac{K_f}{\sum_{g \in \mathcal{G}} K_g}, \quad (10)$$

so that each plant is initially asked to supply the same fraction of its rated output. Equation (10) is an allocation key, the true mix is decided inside the plant-level dispatcher described next.

Merit-order dispatch inside the firm: Each generator receives $Q_{f,t}^{\text{plan}}$. All power recipes r attached to the firm are ranked by their short-run marginal cost $c_r \equiv \text{fuel} + O\&M + \text{carbon taxes}$. Capital cost is ignored since the commitment horizon is one step. The dispatcher walks the merit order and assigns output

$$q_{r,f,t} = \min\{k_{r,f} c_{f,r,t}, q_{r,f,t}^{\max}, Q_{f,t}^{\text{left}}\},$$

- $k_{r,f}$ – capacity of technology r owned by plant f (MW);
- $c_{f,r,t}$ – stochastic capacity factor (drawn in different ranges from $[0.25, 0.95]$ for wind/solar, 1 for dispatchable thermal units);
- $q_{r,f,t}^{\max}$ – Leontief maximum allowed by labor, fuel, and maintenance inventories;
- $Q_{f,t}^{\text{left}}$ – unsatisfied portion of the plant's quota after higher priority recipes have been scheduled.

The loop terminates once $Q_{f,t}^{\text{plan}}$ is reached or all technologies are fully utilised. Total energy offered to the day-ahead auction is $Q_{f,t} = \sum_r q_{r,f,t}$.

Splitting commitment into a system-level quota ($Q_t^{\text{plan}}, Q_{f,t}^{\text{plan}}$) and a plant-level merit order achieves two objectives. First, the ISO takes demand risk off individual firms; second, plants still compete on cost because only cheap recipes secure generation shares when the joint target is smaller than physical potential. Renewables enter the ranking with zero fuel cost but their stochastic $c_{f,r,t}$ may curtail output, creating a dynamic interaction between variable supply and firm-level capacity decisions.

Day-ahead auction: The wholesale electricity market is cleared once per simulation step by a single-price, uniform clearing auction that operates in four logical stages: (i) supply submission, (ii) demand aggregation, (iii) price finding, and (iv) pro-rata allocation. After the commitment routine has determined physical generation $q_{f,t}$, every generator f posts one lot that bundles its entire output at the offer price:

$$p_f = c_{\min,f} (1 + \mu), \quad \mu = 0.10$$

Here $c_{\min,f}$ is the lowest short-run marginal cost among the firm's available power recipes. Collating all offers yields an ordered supply schedule:

$$\{(p_1, s_1), (p_2, s_2), \dots, (p_{N_S}, s_{N_S})\}, \quad p_1 \leq p_2 \leq \dots \leq p_{N_S},$$

where s_i is the kWh quantity attached to price step p_i . Households and non-energy firms submit inelastic day-ahead bids $d_{j,t}$ determined in the demand-formation block. Stacking all bids creates a demand schedule $\{(b_1, d_1), \dots, (b_{N_D}, d_{N_D})\}$ and the step's total demand:

$$D_t^{\text{elec}} = \sum_{j=1}^{N_D} d_j$$

Let the cumulative supply function be:

$$S(p) = \sum_{i: p_i \leq p} s_i$$

The auctioneer searches the increasing supply curve until it meets or exceeds aggregate demand:

$$p^* = \min\{p : S(p) \geq D_t^{\text{elec}}\}.$$

If a policy variable `price_cap` is active, p^* is truncated at $p^{\max}$ to emulate regulatory price caps in stress scenarios. This is currently only used for testing purposes. However, this mechanism could be useful for studying further scenarios in the future.

Let i^* be the first supply step whose cumulative quantity meets demand. All lots with $p_i < p^*$ clear in full. If only a fraction $\lambda \in (0, 1]$ of the marginal lot s_{i^*} is required, the auction divides that lot proportionally among the tied sellers:

$$\lambda = \frac{D_t^{\text{elec}} - S(p_{i^*-1})}{s_{i^*}}$$

Every accepted kWh, buyer side as well as seller side, is settled at the uniform price p^* so that no participant can gain from strategically splitting bids.

The auction is strategy-proof for quantity-taking generators: offering above marginal cost risks non-selection, while offering below incurs a loss if the bid sets the price cap. Marginal-cost pricing guarantees that renewable generation clears before fossil units whenever physical capacity allows, reflecting observed merit-order effects in real-world markets. Because the supply curve is sorted once, and the demand curve only needs aggregation the phase is computationally light even in larger simulations.

For programming convenience the current prototype lets households bid directly in the wholesale auction. This avoids creating an additional retail layer and keeps the demand signal in one place. Future versions should replace these individual bids with an intermediary (retail suppliers or a distribution ISO) to reflect the actual market structure however.

Settlement and feedback: After the call auction clears, four bookkeeping steps close the electricity phase.

(i) Payment flow Deposits move from buyers to generators. A buyer with insufficient liquidity triggers a standard credit routine.

(ii) Price broadcast To ensure subsequent cost-accounting modules observe the market outcome, every generator’s list price field is overwritten: `s.price = p*`.

(iii) Shortage metric Unserved demand is recorded as:

$$\text{shortage}_t = \max\{0, D_t^{\text{elec}} - S(p^*)\}$$

(iv) Adaptive safety margin The parameter `energy_safety_margin` controls how much planned output Q^{plan} exceeds forecast demand (cf. Eq. (10)). When $\text{shortage}_t > 0$ the margin is raised by +25 % (capped at 50 %); when demand is fully met it decays by 10 % per step, down to a floor of 1 %. This simple integral controller nudges the system toward a quasi-stable reserve level without hard-coding exogenous capacity.

(v) Optional production subsidy For experiments that require reserve generation even when the market price fails to cover costs, the scenario parameter `energy_subsidy_per_kwh` transfers a government rebate σ [€/kWh] to each generator:

$$\text{gov_subsidy}_{f,t} = \sigma Q_{f,t}$$

Setting $\sigma > 0$ mimics capacity-payment schemes. This again, is used primarily for testing and investigation purposes at the moment, controlling for setups where the price of energy might have been structurally too high for a not immediately clear reason. A more generalized subsidy system for critical production branches could be another future extension however, and might even stabilize currently problematic simulation runs to some degree.

The modeled electricity market structure, incorporating uniform-price auctions, merit-order dis-

patch based on marginal costs, and integrated demand management, mirrors the design and operational principles of contemporary European day-ahead markets, as implemented under the EU Target Model framework [Gómez et al., 2019].

3.2.4 Additional production mechanisms

The generic loops described reproduce the headline dynamics of the real-side economy, yet some smaller mechanisms are active in the background.

Two auxiliary hour blocks are added to a firm’s wage bill before the first unit of output is even attempted:

- 1) **Fixed hours** A constant amount H^{fix} is charged *every* step in which the firm has at least one employee,

$$H^{\text{fix}} = H_{\text{step}} \phi^{\text{fix}}, \quad 0 < \phi^{\text{fix}} \ll 1.$$

With the baseline $\phi^{\text{fix}} = 0.05$ and $H_{\text{step}} = 160$ h each worker is paid for an extra 8 h of supervision, cleaning, or paperwork per month even if the production quantity $Q_{f,t}$ turns out to be zero.

- 2) **Start-up hours** Whenever the firm has been idle for $\tau \geq 1$ consecutive steps it must first “warm-up” the production line. A one-off block

$$H^{\text{start}} = H_{\text{step}} \phi^{\text{start}}, \quad \phi^{\text{start}} = 0.075,$$

is added on the *first* step in which $Q_{f,t}^{\text{plan}} > 0$ again, after which the idle counter is reset to 0. The start-up charge therefore penalizes stop-and-go operation but vanishes in continuous production.

If $L_{f,t}$ is the head-count at the end of the labor phase the pay-as-you-go wage bill that must be financed before any sales revenue materializes is

$$W_{f,t}^{\text{pre}} = w_{f,t} L_{f,t} \left(H^{\text{fix}} + \mathbf{1}_{\{\tau > 0\}} H^{\text{start}} \right),$$

where $w_{f,t}$ is the hourly wage and $\mathbf{1}_{\{\tau > 0\}}$ is an indicator that equals 1 only on the first production step after an idle spell. Because $W_{f,t}^{\text{pre}}$ is strictly positive as long as the firm has employees, aggregate wage payments exhibit a floor that cushions the income effect of sharp output contractions and dampens volatility in the consumption block of the model. The two parameters ϕ^{fix} and ϕ^{start} can be calibrated independently to empirical ratios of overhead labor and restart downtime in specific industries.

Fallback procurement: If the spot market holds no inventory a firm can still buy directly from any tier-0 producer at its list price. This ‘standing contract’ path prevents complete dead-locks when an upstream sector has capacity but no finished goods on hand.

Upstream-shortage feedback: Input gaps registered this step are aggregated into the dictionary `upstream_shortage`. In the next labor phase that signal is added to the backlog term of Eq. (3), nudging suppliers to raise their targets and propagating supply-chain shocks forward along the network.

3.3 Household–consumption phase

The goods market clears after firms have finished production but before new investment is considered, so that household purchases reflect the most recent output and price signals. The phase is executed once per simulation step and transforms deposits held by households into sales revenue for

firms, nominal GDP, and an updated consumer-price index. The consumption module is inspired by microeconomic foundations of consumer behavior [Deaton and Muellbauer, 1980], with stylized utility-based allocation rules that separate subsistence and discretionary spending. A decile-based marginal propensity to consume allows for non-linear Engel-curve effects, ensuring higher-income households consume proportionally less.

High-level sequence

- C1)** Budget assessment for every household (deposits minus mandatory loan payments)
- C2)** Subsistence consumption out of a fixed basic bundle
- C3)** Discretionary consumption allocated by utility weights
- C4)** Soft-max firm choice and physical trade execution
- C5)** Deposit transfers, inventory updates, and bookkeeping
- C6)** Price-index, GDP, and unmet-demand statistics

Budget assessment Before consumption starts each household accounts for any eventual scheduled debt service. Based on $dep_{h,t}$, the currently available deposits before repayment and $L_{h,t}$ the mandatory loan installment due this step. The disposable budget that enters the allocation rules below is:

$$B_{h,t} = \max\{0, dep_{h,t} - L_{h,t}\}.$$

If outstanding deposits do not cover $L_{h,t}$ the household's disposable budget is clipped to zero.

Budget allocation and basket construction At the start of the consumption phase every household h holds disposable deposits $B_{h,t}$. A wealth-decile-specific marginal propensity to consume

$$\lambda_h = \text{MPC}(\text{decile}_h), \quad 0 < \lambda_h \leq 1,$$

limits total outlays. The effective consumption budget is

$$B_{h,t}^{\text{cons}} = \lambda_h B_{h,t},$$

while the residual $(1 - \lambda_h)B_{h,t}$ is saved and carried forward. Within $B_{h,t}^{\text{cons}}$ a constant share $\phi_{\text{sub}} = 0.5$ is marked for subsistence spending:

$$B_{h,t}^{\text{sub}} = \phi_{\text{sub}} \lambda_h B_{h,t}.$$

At model initialization a subsistence list $\mathcal{G}^{\text{sub}}$, common to all households, is drawn from tier-2 consumer goods as a random half-sample $|\mathcal{G}^{\text{sub}}| \approx 0.5 |\text{tier-2}|$. Households nevertheless differ in how strongly they value each basic item: a weight $u_{g,h} \sim \mathcal{U}[0.8, 1.2]$ is drawn independently for every pair (g, h) . Subsistence spending allocates the subsistence budget proportionally to these weights and inversely to current prices,

$$\alpha_{g,h,t} = \frac{u_{g,h} p_{g,t}^{\min-1}}{\sum_{k \in \mathcal{G}^{\text{sub}}} u_{k,h} p_{k,t}^{\min-1}}, \quad g \in \mathcal{G}^{\text{sub}},$$

so that

$$q_{g,h,t}^{\text{sub}} = \frac{\alpha_{g,h,t} B_{h,t}^{\text{sub}}}{p_{g,t}^{\text{min}}}, \quad \sum_g p_{g,t}^{\text{min}} q_{g,h,t}^{\text{sub}} = B_{h,t}^{\text{sub}}$$

The remaining discretionary budget

$$B_{h,t}^{\text{disc}} = (1 - \phi_{\text{sub}}) \lambda_h B_{h,t},$$

is allocated over the complementary basket $\mathcal{G}^{\text{disc}} = \text{tier-2} \setminus \mathcal{G}^{\text{sub}}$, using the same rule. Both routines use the same soft-max firm-choice outlined next. Because the saving rate $(1 - \lambda_h)$ scales the entire budget, richer households cut back proportionally on both tiers. This serves as a simplification for now until a richer micro-foundation with more explicit dynamics for essential consumption as well as richer job and wage dynamics are implemented.

Firm choice Within each good category households face a discrete set of firms that list positive prices. The probability of selecting firm f is a soft-max over the inverse of the posted prices,

$$P_{h \rightarrow f} = \frac{\exp[(p_f^{-1} - p_{\max}^{-1})/(T p_{\max}^{-1})]}{\sum_j \exp[(p_j^{-1} - p_{\max}^{-1})/(T p_{\max}^{-1})]}, \quad T_{\text{cons}} = 0.3 \text{ (baseline)}, \quad (11)$$

where $p_{\max}$ is the highest price in the candidate set. A low temperature ($T \rightarrow 0$) yields almost-greedy, price-minimizing behavior; larger T smooths the choice probabilities.

Trades are attempted until either the physical need, the monetary allocation, or the supplier's inventory is exhausted. A market abstraction removes the corresponding goods from the chosen firm in one batch and transfers ownership to the household. If a household cannot afford even one unit at the selected supplier the loop terminates, and the unspent allocation is recorded as unsatisfied demand. The same happens if no supplier with positive stock exists. In the next planning round these counts are injected into the backlog term of the standard firm-demand rule, transmitting retail bottlenecks upstream.

Macro metrics Nominal consumption GDP equals the sum of all payments executed this step. A Paasche price index is computed from the transaction set $\{(p_i, q_i)\}$:

$$P_t = \frac{\sum_i p_i q_i}{\sum_i q_i},$$

and real GDP is $Y_t^{\text{real}} = Y_t^{\text{nom}}/P_t$. The same transaction list updates a rolling sales-price history for each firm. This history drives future recipe-cost estimates on the production side.

Household infrastructure demand In addition to market goods, every household consumes a small amount of public infrastructure service each step. The baseline quantity scales linearly with population,

$$D_t^{\text{hh,infra}} = \iota_{\text{hh}} N_{\text{hh}}, \quad \iota_{\text{hh}} = \text{SI} \bar{\iota}_{\text{hh}}, \quad (12)$$

where $\bar{\iota}_{\text{hh}} \approx 0.01$ is the reference requirement and $\text{SI} \in (0, \infty)$ is the service-intensity calibration parameter. Since infrastructure services are provided as a public service however, this consumption does not affect the households budget.

3.4 Investment and Capital Accumulation

Investment decisions in the model are governed by firm-level signals that reflect both capacity constraints and economic incentives. The implementation draws on established micro- and macroe-

conomic literature that links capital accumulation to expected profitability, asset valuations, and financing conditions [Hall and Jorgenson, 1967, Tobin, 1969]. Trigger mechanisms based on utilization, profit margins, and Tobin’s q determine whether a firm attempts to expand capacity. The resulting investment gaps are subject to diminishing returns. All ordered kits enter a multi-step pipeline with stochastic lead times and installation lags, allowing for delay structures and temporal smoothing of aggregate capital formation. In the power sector, investment decisions additionally incorporate leveled cost minimization across competing energy technologies [International Energy Agency, 2020].

High-level sequence

- I1) Process pipeline** – deliver or install kits whose time-stamps fall due.
- I2) Trigger check** – evaluate utilization, profitability, and Tobin- q signals.
- I3) Order sizing** – translate the active trigger into a capacity gap and determine the integer number of kits.
- I4) Financing & procurement** – choose between in-house build, credit-financed purchase, or energy-sector technology switch.
- I5) Stamp order** – attach lead time L and installation lag ℓ and enqueue the kit in the firm’s investment pipeline.

Each step is executed for every active firm, with specialized branches based on firm type in the respective high-level investment function.

Process pipeline Every firm maintains an `investment_pipeline` containing kit orders with delivery and installation time-stamps (t_{deliv} , t_{inst}). Status changes follow

$$\text{ordered} \xrightarrow{t \geq t_{\text{deliv}}} \text{delivered} \xrightarrow{t \geq t_{\text{inst}}} \text{installed},$$

and an installed kit adds its name-plate capacity κ_{kit} to K_f or to the respective `capacity_by_tech` for firms in the power sector.

Trigger check Before placing a new order the firm consults three independent signals that proxy capacity stress, profitability, and asset-market valuation. Let

$$u_f = \frac{Q_{f,t}}{\bar{Q}_{f,t}}, \quad m_f = \frac{\pi_{f,t}}{R_{f,t}}, \quad q_f = \frac{\pi_{f,t}/K_{f,t}}{c_u},$$

denote, the realized utilization rate, the operating profit margin, and Tobin’s q . With i the loan rate faced by the firm, δ the economic depreciation rate, τ the profit-tax rate, and $\dot{p}/p$ the expected capital-gain term, the user cost of capital is

$$c_u = i + \delta - \tau + (1 - \tau) \dot{p}/p.$$

- (i) **Utilization trigger:** Capacity is deemed tight when $u_f > u^*$ with a baseline $u^* = 0.8$. The smoothed reference output $\bar{Q}_{f,t}$ is an exponentially weighted moving average of the last twelve realized volumes, $\bar{Q}_t = \gamma Q_{t-1} + (1 - \gamma) \bar{Q}_{t-1}$ with $\gamma = 0.2$.
- (ii) **Profit-margin trigger:** Even if utilization is moderate, a high mark-up can still justify expansion. The firm therefore orders when $m_f > m^{\min}$ with $m^{\min} = 0.05$.

- (iii) **Tobin-q trigger:** The rational-expectations benchmark is met when $q_f > 1$, i.e. the present value of the expected earnings stream exceeds replacement cost. To avoid false positives from short-run noise, $\pi_{f,t}$ in the numerator is taken as a rolling 12-month sum.

A Boolean indicator tells the procurement routine whether to proceed to the sizing step. The default thresholds can be overwritten in scenario files to emulate credit crunches ($u^* \uparrow$), investment booms ($m^{\min} \downarrow$), or periods of over-valued equity ($q \downarrow$). These parameters generally could also be exposed to and modulated by a model extension in the future, resulting in more adaptive behavior on a global and firm level.

Gap calculation and order size Each trigger produces a scalar σ_f (u_f , m_f or q_f) that needs to be translated into a percentage expansion target $g(\sigma_f) \in [0, 1]$. Close to the threshold the map is linear, far above it a square-root branch ensures diminishing incremental appetite:

$$g(\sigma_f) = \begin{cases} \phi_{\text{adj}}(\sigma_f/\sigma^* - 1), & \sigma_f/\sigma^* \leq 1 + \theta_g, \\ \phi_{\text{adj}}(\theta_g + \sqrt{\sigma_f/\sigma^*} - 1), & \text{otherwise,} \end{cases} \quad \theta_g = 0.25.$$

With $\phi_{\text{adj}} = 0.2$ a utilization of 85% (only 6% above the 80% hurdle) asks for a modest $0.2 \times (0.85/0.80 - 1) = 0.0125$ or 1.25% capacity addition. A surge to 100% utilization still scales orders but less than proportionally because of the square-root term. Multiplying the percentage by the active stock yields the raw physical gap:

$$\Delta K_f^{\text{raw}} = g(\sigma_f) K_f$$

Large projects face co-ordination and construction frictions. The model penalizes gaps that exceed a fraction θ_{adj} of the base stock with a quadratic surcharge parameterized by χ :

$$\Delta K_f = \frac{\Delta K_f^{\text{raw}}}{1 + \chi \left(\frac{\Delta K_f^{\text{raw}}}{K_f} - \theta_{\text{adj}} \right)^2}, \quad (\theta_{\text{adj}} = 0.30, \chi = 1).$$

Attempting to grow equipment by e.g. 50% in one step— $\Delta K^{\text{raw}}/K_f = 0.5$ shrinks to an effective $0.5/[1 + 1(0.5 - 0.3)^2] \simeq 0.42$. Finally the physical gap is divided by the name-plate capacity κ_{kit} of one investment kit and rounded to the nearest integer:

$$n_f = \left\lfloor \Delta K_f / \kappa_{\text{kit}} \right\rfloor, \quad n_f \geq 0.$$

A minimum of one kit is enforced only when the trigger is active (σ_f above threshold); otherwise $n_f = 0$ and the firm skips ordering this step.

Financing and procurement Once the investment decision is set, the firm must answer two practical questions: (i) who supplies the kits? and how are they paid for? The model recognizes three distinct channels.

1. **Self-investment** Firms whose output good are `Investment_Goods` can use those internally. No market purchase is needed, instead an opportunity-cost transfer is booked:

$$C_{\text{kit}}^{\text{opp}} = \pi_{\text{kit}} - c_{\text{unit}},$$

where π_{kit} is the going market price and c_{unit} the firm's own variable cost.

2. **Credit-financed market purchase** For all other incumbents the default path is to buy kits from the pool of investment-goods firms.

- (i) *Price* — Each kit costs the seller’s current list price π_{kit} .
- (ii) *Deposit check* — If deposits exceed the invoice, the kit is paid in directly. Otherwise the firm requests a bank loan.
- (iii) *Free-cash-flow cap* — A loan is issued only if the projected annual debt service (interest + principal) stays below a fraction ψ_{fcf} of last-year’s operating profit:

$$\frac{\text{ADS}_{\text{new loan}}}{\sum_{s=t-11}^t \max(\pi_{f,s}, 0)} \leq \psi_{\text{fcf}}, \quad \psi_{\text{fcf}} = 0.40.$$

- (iv) *Amortisation* — Granted loans are recorded in the finance module and amortised over 60 steps (5 years) at the current bank rate i_b .

3. Energy-sector technology choice Power generators face a multi-technology menu. Whenever they place an order via channel 2 they assign each kit to the technology that minimizes a levelized cost of electricity (LCOE):

$$\text{LCOE}_r = p_{\text{fuel}} \beta_r + w_f \lambda_r + (i + \delta) k_r,$$

combining fuel, labor, and capital charges evaluated at current prices. The installed capacity is stored in the dictionary `capacity_by_tech[tech_tag]` so that dispatch and emissions accounting can track the evolving composition.

If aggregate kit inventory is smaller than the requested n_f , the order is partially filled; the unmet quantity is counted in `unmet_investment_demand` and feeds into backlog, encouraging the specialized producers to scale up.

Lead-time and installation-lag stamping An order that passes the financing check is not available instantaneously. Instead the model attaches two stochastic clock variables before the kit enters the firms `investment_pipeline`:

L **Lead time** — months between the purchase contract and physical delivery,

ℓ **Installation lag** — months required for foundation, grid-connection, commissioning, etc. after the kit is on-site.

Both are drawn from uniform distributions whose bounds depend on sector and, for energy firms, on project scale:

$$L \sim \mathcal{U}[L_{\min}(s), L_{\max}(s)], \quad \ell \sim \mathcal{U}[\ell_{\min}(s), \ell_{\max}(s)],$$

where the scale index $s \in \{\text{small}, \text{medium}, \text{large}, \text{XL}\}$ is chosen from the current plant size. Default monthly ranges, configured in the tuning YAML, are:

Sector / scale	$L_{\min}$	$L_{\max}$	$\ell_{\min}$	$\ell_{\max}$
Manufacturing (all)	2	4	0	1
Power, small (<50 MW)	2	4	1	2
Power, medium (<200 MW)	6	12	2	4
Power, large (<500 MW)	12	24	3	6
Power, XL (≥ 500 MW)	24	36	4	8

Table 1: Head-count and per-worker labor coefficients by sector and scale.

In the monthly time step of the simulation these values translate 1-to-1 into step counts. A kit ordered in step t therefore becomes operational only at

$$t_{\text{online}} = t + L + \ell.$$

Lead-time uncertainty creates a pipeline of capacity, which (i) tempers investment booms by introducing construction congestion, (ii) allows policy shocks, e.g. a carbon price hike, to shift the technology mix of future rather than current capacity, and (iii) generates realistic lagged responses in aggregate capital and electricity supply. Because lead-time ranges are scenario-tunable, the model can replicate fast-track programmes (compress L_{max}) or supply-chain bottlenecks (stretch both L and ℓ). These too are currently exogenous parameters however that should ideally become dynamic in future iteration.

3.5 Market Entry of New Firms

The model includes a market entry module to complement its bankruptcy routine, handled by the finance module. Without such a mechanism the economy would eventually dead-lock during run-time since successive firm failures would sooner or later erode every critical supply node. Entry therefore fulfills a dual role: it restores indispensable sectors and is addressing potential bottlenecks. Entry is evaluated once per simulation step, immediately after the investment phase, and creates firm agents, funds their start-up balance sheet, and registers it with the goods market.

ME-sequence The entry routine executes the following ordered tests in every step. If a test succeeds it triggers the spawning logic described further below, otherwise the next test is evaluated.

- ME1) Power-sector trigger** Add a new generator when demand outstrips buffered supply, when electricity prices persistently exceed the lowest LCOE by at least 15%, or when average utilisation of electricity users falls below 60% because of rationing.
- ME2) Critical-input safeguard** If no surviving producer exists for a critical good (electricity, coal, gas, oil) a replacement firm is spawned instantly, bypassing all finer triggers.
- ME3) Extractor trigger** For each fossil fuel compare an up-to-date demand signal with a smoothed supply signal. A new extractor is created when demand exceeds supply by more than 10%.
- ME4) Generic-goods trigger** For every other tier-2 consumer good a battery of weak heuristics—persistent sell-outs, excessive market price, elevated producer margins, downstream utilisation squeeze, high incumbent utilisation—are evaluated. Any one of them is sufficient to justify entry.
- ME5) Spawn firm** When the first satisfied condition above returns `TRUE`, `spawn_firm()` is called to instantiate the entrant.

Trigger definitions: Entry relies on an intentionally simplified rule set. Firms appear only when evaluated signals point to sustained excess demand or critical input collapse. The approach keeps the algorithmic burden linear in the number of goods, yet it inevitably sacrifices the nuance a fully fledged optimization could provide. In particular, the present logic (i) has no forward expectation, reacting only after imbalances are observable; (ii) treats each good in isolation, ignoring cross-price and cross-input elasticities; and (iii) cannot distinguish transitory price noise from structural scarcity in very short simulations. This module will need a more general overhaul. Thresholds listed

below correspond to the baseline scenario and can be tightened or relaxed to emulate, respectively, a more laissez-faire or a more activist industrial environment.

(a) Power sector (entry of new generators)

- (1) *Demand-buffer breach* A generator is added whenever system demand D_t exceeds buffered supply by more than two percent, formalised as $D_t > \alpha_{\text{buf}} S_t$ with $\alpha_{\text{buf}} = 1.02$. The small margin prevents tight balancing intervals without waiting for blackouts.
- (2) *Price-LCOE gap* If the four-step moving-average wholesale price $\bar{p}_{t,4}$ surpasses the cheapest levelised cost of electricity by at least fifteen per cent ($\bar{p}_{t,4} > 1.15 \min_r \text{LCOE}_r$), the model interprets the spread as an investment opportunity. Because the rule uses the minimum rather than the mean LCOE it is biased towards over-entry when the technology mix is diverse. Which is an artifact of a previous implementation, where electricity was structurally undersupplied.
- (3) *Downstream rationing* Persistent utilization below sixty percent among electricity consumers indicates forced curtailment and triggers entry even when the previous two criteria are silent. The catch-all nature of this rule can lead to overshooting during short-lived demand spikes.

All three tests rely on backward-looking moving averages. Sudden regulatory shocks (e.g. carbon price hikes) may thus lead to delayed renewable build-out.

(b) Fossil extractors (coal, gas, oil)

- (1) *Fuel gap*. For each fuel g a new extractor is spawned when the purchase-based demand signal $\mathcal{D}_g$ exceeds a smoothed supply signal $\bar{\mathcal{S}}_g$ by more than ten percent: $\mathcal{D}_g > 1.10 \bar{\mathcal{S}}_g$. The gap uses a one-sided filter, so short-term hoarding behaviour can misfire the trigger.

Extractor entry ignores remaining reserves, capital intensity and permitting delays; it simply assumes any demand surplus can be met by drilling another well. This is reasonable for the current stylized fuel layer but unsuitable for detailed depletion studies.

(c) Other Firms

- (1) *Persistent sell-outs* Incumbent firms whose sales-to-production ratio exceeds 0.90 for three consecutive steps are deemed unable to build inventories, signaling latent excess demand.
- (2) *Price pressure* A four-step average market price more than twenty-five percent above the contemporaneous median incumbent price triggers entry. The median guards against outliers from distressed sellers.
- (3) *High producer margins* Mean operating margins above ten percent over the same window also signal an attractive niche.
- (4) *Downstream bottleneck* Average utilization of firms *using* the good below 0.60 suggests input scarcity and calls for additional suppliers.
- (5) *Producer congestion* Utilization of incumbents above 0.85 captures the symmetric case of capacity saturation at the supply side.

The heuristics are evaluated in isolation. If two countervailing signals, e.g. high price but low margin, co-exist, the first in the list prevails. Moreover, only one entrant per good per step is allowed, which can delay convergence in goods with extremely small kit sizes. This behavior prevents cascades of simultaneous spawns however.

Once a trigger fires the model constructs a new firm agent. The machinery is intentionally lightweight: it equips the newcomer for a single step, but leaves long-run survival to ordinary profit-and-loss dynamics.

Every entrant receives a block of tangible capital in the form of investment kits. For power companies the number of kits is derived from the estimated capacity gap,

$$n_{\text{kit}} = \left\lceil 0.8 \frac{\text{Gap}_{\text{MW}}}{\kappa_{\text{kit}}} \right\rceil,$$

so about 80 % of the short-fall is closed in the next construction cycle; the residual keeps prices informative for incumbent expansion. For all other sectors the model uses a pragmatic heuristic, one kit for every five workers in the minimum crew. Kits are bought from the investment-goods sector at the incumbent average price; if no kits are available the failed attempt increments `unmet_investment_demand`, nudging kit producers to scale up later.

Besides kits the entrant needs cash to pay wages before any sales revenue materializes. Deposits therefore cover the kit invoice plus one step of fixed and start-up labor hours for the minimum crew. The default funding mode is credit-based: the firm borrows from its designated commercial bank. All recipes whose `output_id` matches the good type to be produced are attached, providing an immediate choice set. The active recipe for the very first production step is still chosen by the standard cost-minimization rule.

Because the entrant receives just enough capital to operate but no guarantee of future credit, the mechanism acts as a weak stabilizer. Its current systemic purpose is to prevent supply-chain collapse, and to ease persistent bottlenecks, while not suppressing market signals entirely: price spikes and inventory shortages must actually occur before entry is triggered, and poorly chosen entrants can still go bankrupt. The module lacks more dynamic behavior however. Computationally speaking the routine is lightweight in return. It operates on sector-level aggregates and spawns at most one firm per good type and step. Runtime therefore grows only linearly with the number of goods.

3.6 Fiscal Policy

The government acts in one consolidated fiscal phase executed after production and consumption are finalized but before firms decide on new investment. All public cash flows go through a `FiscalTracker`, which records revenues, outlays, and bond issues so that ex-post statistics never miss a transaction.

High-level sequence

- FP1) Collect taxes**—progressive income, profit, and scenario-dependent carbon.
- FP2) Pay transfers**—unemployment benefits and, if the carbon module is active, lump-sum rebates.
- FP3) Aggregate outlays**—sum the infrastructure purchases that public firms have already drawn during the labour, production, or investment stages.
- FP4) Close the fiscal balance**—compute the surplus/deficit and enforce the 60 % debt-to-GDP ceiling.

Tax collection Households are ranked by financial net worth and placed into ten equal-sized wealth groups. A gradual income-tax schedule listed in Table 2 is applied based on decile position

afterwards. Rates are broadly oriented around values reported by OECD countries [OECD, 2025]. A five-step ramp-in $\tau_{\text{eff}}^Y(t) = \min(1, t/5) \tau^Y$ spreads the initial cash-flow hit:

$$T_h^Y = \tau^Y(d(h)) [w_h^{\text{lab}} + r_h^{\text{dep}}],$$

where w_h^{lab} is current labor income and r_h^{dep} the interest on any eventual deposits.

Wealth decile d	1	1	2	3	4	5	6	7	8	10
Income-tax rate $\tau^Y(d)$ (%)	0	5	10	15	25	30	35	40	45	55

Table 2: Progressive household income-tax schedule used in the baseline scenario.

For firms a flat profit tax rate rate $\tau^\Pi = 25\%$ is applied. This as well is broadly in line with rates observed across OECD countries [OECD, 2024]. The sum of wage, energy, intermediate-input and loan-servicing costs is reported for the step, as total costs which is subtracted from gross revenue. Loss-making firms face zero liability, while highly leveraged or capital-intensive firms with temporarily low cash flow are not penalized.

Unemployment Benefits Non-employed households receive a replacement benefit b_h based on their last wage payment w_h^{last} and capped by $\bar{w}_t$, the average hourly wage in step t

$$b_h = \min[\psi_{\max} \bar{w}_t, \max(\psi_{\min} \bar{w}_t, \rho_{\text{rep}} w_h^{\text{last}})],$$

with replacement rate $\rho_{\text{rep}} = 0.6$, floor $\psi_{\min} = 0.2$ and ceiling $\psi_{\max} = 3.0$ expressed as multiples of $\bar{w}_t$. The benefit is linearly scaled down once a household's financial net worth W_h^{nw} rises above ϕ_W times the population mean $\bar{W}_t^{\text{nw}}$:

$$b_h^{\text{net}} = \beta_h b_h, \quad \beta_h = \begin{cases} 1, & W_h^{\text{nw}} \leq \phi_W \bar{W}_t^{\text{nw}}, \\ \max[0.10, 1 - 0.90 \frac{W_h^{\text{nw}} - \phi_W \bar{W}_t^{\text{nw}}}{\phi_W \bar{W}_t^{\text{nw}}}], & W_h^{\text{nw}} > \phi_W \bar{W}_t^{\text{nw}} \end{cases}$$

A household at the threshold ($W_h^{\text{nw}} = \phi_W \bar{W}_t^{\text{nw}}$) receives the full benefit ($\beta_h = 1$).

Government-funded outlays Infrastructure is currently the sole mandatory public item in the baseline configuration. The `InfraServiceFirm` bills the Treasury on the spot during the production phase, and the fiscal phase merely aggregates those invoices into `infrastructure_spending`. No new payment is initiated here, keeping this as a pure accounting step.

Deficit management The government enforces a simple Maastricht-style spending ceiling $d_t/\text{GDP}_t \leq 0.60$, reflecting the EU Stability and Growth Pact target [European Council, 2012]. When deficit violates the limit, benefits and rebates are scaled down proportionally, spending in public firms gets blocked. As this is mainly an accounting based flag, the code allows for overstepping the ceiling for one step and forces compliance in the following steps until the debt-to-GDP falls under the threshold again. Tax rates are not altered as a response.

3.7 Financial Layer

The financial layer of the model implements a full stock-flow-consistent banking system, integrating central bank operations, commercial banking, government debt, and non-financial agent behavior. Its purpose is to capture how money and credit are created, circulated, and extinguished, and how

interest rates and fiscal instruments interact to shape aggregate demand, financial stability, and sectoral balance-sheets.

3.7.1 Financial Agents

The model distinguishes three core financial agent types and two classes of private agents:

- **Central Bank (CB):** Issues reserves, sets policy rates via a Taylor rule, manages the government's and the commercial banks accounts.
- **Commercial Banks:** Manages household and firm accounts, while holding an account at the central bank. Creates deposits via balance sheet extensions and holds a minimum reserve of 1% with the CB [European Central Bank, 2011]
- **Government:** Issues bonds to finance deficits, levies taxes, and makes transfers and spending payments via its CB account.
- **Households:** Hold deposits, take out loans, receive wages and transfers, and pay taxes.
- **Firms:** Borrow for working capital and investment, hold deposits for operations, and pay wages and taxes.

Setting up the agents this way results in a two-tiered banking system (Figure 3), with distinct reserve-based and deposit-based monetary cycles. Both the CB and commercial banks create funds via balance sheet extensions, primarily when giving out loans to recipients, and since they only are able to hold accounts for specific types of agents the two layers only interact via the commercial banks. A CB loan to a commercial bank creates new reserves, a commercial bank loan to a non-bank creates new deposits. Reserves are central-bank liabilities held only by commercial banks and the government, used for settlement and to fulfill minimum reserve requirements. Deposits are commercial bank liabilities held by firms and households for transactions [McLeay et al., 2014]. As funds are created via balance sheet extensions during loan creation, loan repayments lead to a contraction of balance sheets destroying those funds again. Interbank transfers are handled with reserves without altering the total money stock.

Government fiscal operations bridge the two accounting tiers in a stock-flow-consistent manner: when the government's CB account is insufficient for some spending operation, it issues bonds, transferring reserves from a buying commercial banks CB account into its own CB account. Coupon payments and maturities reverse that flow back. Government spending with a firm or household agent as a target transfers reserves to the recipients commercial bank, which in turn creates deposits for the recipient in the same amount. Tax payments reverse that same flow.

There currently are no bond auction and secondary market systems implemented, the specific commercial bank buying a bond is determined randomly at the moment, other agents do not buy these bonds on a secondary market yet. The general infrastructure for these subsystems is supported by the model already however and should be added as a module in a future iteration for enhanced realism. All monetary flows are recorded on agent balance sheets, managed by their respective bank agent. In baseline and carbon tax scenarios government spending is limited with a hard cap to 60% of current GDP, representing EUs Stability and Growth Pact rules.

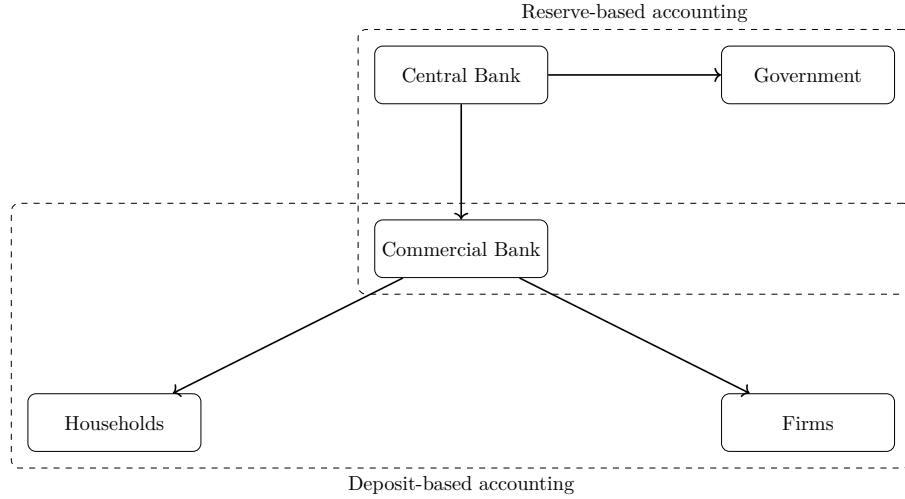


Figure 3: Hierarchy of accounts and the division between reserve-based (top) and deposit-based (bottom) accounting.

3.7.2 Monetary Policy and the Stylized Taylor Rule

The model utilizes a discrete-time Taylor rule [Taylor, 1993] to update the CB's policy rate each simulation step (monthly), targeting both inflation and output gaps. The rule is:

$$r_t = r^* + \pi_t + \phi_\pi (\pi_t - \pi^*) + \phi_y (y_t - \bar{y}),$$

where r_t is the nominal policy rate (CB loan rate) at step t ; $r^* = 0.02$ is the neutral real rate, representing long-run equilibrium real costs of funds; π_t is the annualized inflation rate, computed from CPI (or price index) changes over the past four steps; $\pi^* = 0.02$ is the central bank's inflation target (2% p.a.); y_t is the log of real GDP at step t ; and $\bar{y}$ is the log of potential output, calculated as a 4-step moving average of past GDP. The coefficients $\phi_\pi = 1.5$ and $\phi_y = 0.5$ weight the inflation and output gaps respectively, consistent with standard policy prescriptions that recommend a strong reaction to inflation deviations and a more moderate response to cyclical fluctuations.

To avoid unrealistically abrupt rate swings, first-order smoothing is applied:

$$r_t = (1 - \lambda) r_{t-1} + \lambda r_t^{\text{target}}, \quad \lambda = 0.25,$$

where r_t^{target} is the unsmoothed Taylor-rule rate and λ (25 %) controls the adjustment speed. The CB then sets its deposit rate at $r_t - 0.0015$ to discourage arbitrage between reserves and deposits.

3.7.3 Banking Operations

Commercial banks in the model dynamically set their deposit and loan interest rates based on central bank policy rates and internal strategy parameters. Each bank sets its deposit rate slightly above the central bank deposit rate by adding a small random spread:

$$r_{\text{bank}}^{\text{dep}} = r_{\text{dep}}^{\text{CB}} + \varepsilon_{\text{dep}}, \quad \varepsilon_{\text{dep}} \sim U(0.1\%, 0.3\%).$$

Loan rates are then calculated by applying a minimum margin and an additional small spread to the deposit rate:

$$r_{\text{base}}^{\text{loan}} = r_{\text{bank}}^{\text{dep}} + \text{margin}_{\text{min}}(2\%) + \varepsilon_{\text{loan}}, \quad \varepsilon_{\text{loan}} \sim U(0\%, 0.5\%).$$

The final loan interest rate includes a borrower-specific risk premium determined by a credit score:

$$r^{\text{loan}} = r_{\text{base}}^{\text{loan}} + (1 - \text{score}) \cdot \lambda,$$

where λ is a scaling factor for the risk premium (typically 0.15).

Credit scores are computed using a rule-based function that evaluates borrower attributes. For households, employment status and net worth affect the score; for firms, profitability and capacity utilization play a role. Scores range from 0 (high risk) to 1 (low risk), with penalties for financial distress and bonuses for sound performance.

Loan approval depends on two criteria: the borrower's score must exceed a minimum threshold (e.g., 0.6), and the risk-adjusted margin must be sufficiently profitable for the bank. Approved loans are issued immediately by creating matching deposits, reflecting standard endogenous money principles (financing through money creation) [Jakab and Kumhof, 2015]. Banks then manage their balance sheets by accruing interest and paying interest on deposits. These mechanisms are handled at the end of each step.

Commercial bank behavior in regards to interest and deposit rate setting still relies on heuristics that would have to be addressed in the future. Currently their overall profit seeking behavior is underdeveloped, leading in some configurations to exponential runaway dynamics in the sectoral balances.

3.7.4 Bankruptcy and Default Handling

The model detects financial distress through a rule-based default mechanism. An agent is considered in default if its financial net worth is negative, it holds less than one unit of liquid assets and is unable to refinance through a new loan.

Once defaulting agents are identified, a bankruptcy routine is triggered. For firms, this entails full removal from the simulation (unless exempt, e.g., infrastructure providers). Prior to exit, their real assets and input inventories are equally redistributed to surviving firms within the same sector, preserving productive capacity.

Households, by contrast, are not removed. Instead, they are flagged as bankrupt and immediately lose further borrowing eligibility. They are marked with a `recently_defaulted` flag, which would allow for explicit default recovery logic in a future extension.

Default checks and bankruptcy handling are executed at the end of each simulation step, following financial settlement. This timing ensures that stock-flow consistency is preserved and bankruptcies are promptly resolved.

3.8 Scenario Modules: Carbon Pricing and Job Guarantee

Policy experiments in the simulator are toggled through a `ScenarioFlags` at start-up. Three scenarios are currently structured, a carbon tax scenario, a job-guarantee scenario and a joint execution, activating both modules.

3.8.1 Carbon-pricing module

In this model a tax-and-dividend design is implemented, including a lump-sum redistribution mechanism akin to the carbon dividend approach advocated by Klenert et al. [2018]. After the initial step the scheduler pins the carbon price p_c such that the taxes raised in the next step (1) are equal to θ_{GDP} percent of nominal GDP:

$$p_c(1) = \frac{\theta_{\text{GDP}} \text{GDP}_0}{E_0}, \quad E_0 = \sum_f e_{f,0},$$

with θ_{GDP} set to 1% per default.⁷ After the price follows a user-defined linear path expressed relative to the initial level: $p_c(t) = p_c(1) [1 + \gamma (t/\text{year})]$, where γ is the annual growth rate. The scheduler updates p_c every `step_size` steps (default 12 $\Leftrightarrow$ monthly data).

Each firm records its step emissions based on fuel usage in the active production recipe and the tax is applied, settled and accounted for within the standard costing and taxation routines. Gross revenue, credits and net revenue are logged in the `FiscalTracker`. A share φ_{rebate} of net carbon receipts is returned to households as an equal lump-sum: $b^{\text{ctax}} = \varphi_{\text{rebate}} \Pi_{\text{net}}^C / N$. The remainder flows into the general budget. All carbon flows interact with the usual debt rule in Sect. 3.6, no special fund is created.

3.8.2 Job-guarantee (JG) module

The Job Guarantee (JG) module models a public employer-of-last-resort program inspired by proposals for automatic stabilizers in the tradition of functional finance. In particular, it follows the buffer stock approach laid out by Fullwiler [2007], where a fixed-wage job offer is made to all unemployed individuals. The number of workers employed in the program fluctuates, absorbing idle labor during downturns.

Activating the `job_guarantee` flag adds a ring of productive, public JG firms that offer a standing job to any unemployed worker while producing socially useful outputs. These firms are inheriting from the general firm agent class and therefore show all the same general behavior. They are handled by the same high-level functions, just with modified behavior.

During initialization one JG firm is spawned for every good in the subsistence bundle plus `ELECTRICITY` and `HYDROGEN`. The recipe with the lowest CO_2 intensity for that good is attached, circumventing cost minimizing behavior. JG firms must purchase inputs and invest in capital like any private producer, in contrast to the private producers initialized at setup however, they do not start with any seeded capacity or inputs. Putting them at a disadvantage at the beginning and forcing them to compete for investment kits.

Wages in JG firms are recalculated each step as

$$w_t^{\text{JG}} = \max(\mu w^{\min}, P_{10}(w_{t-5:t}^{\text{priv}})),$$

the larger of a minimum-wage floor ($\mu = 1.05$ by default) and the 10th percentile of private-sector wages over the last five steps. This keeps the JG rate just above the market floor without chasing cyclical spikes. Hiring begins with a unified application round where job-seekers choose between private and JG vacancies via the wage-weighted soft-max introduced earlier. Private firms then hire from their applicant pools, afterwards JG firms take everyone who applied to them and finally absorb any remaining unemployed workers. This sequence guarantees full employment while still letting private firms compete for labor on market terms.

Any input purchases, payroll and investment of JG firms are funded instantly within the respective routines directly. This is the same behavior used for public infrastructure firms. All items are aggregated into `jg_spending` during the fiscal phase, so expenses are traceable. When the JG flag is active the debt-to-GDP ceiling is suspended, reflecting functional-finance arguments that the employer of last resort should not be deficit-constrained until full employment is reached.

At the time of writing initialization of subsistence goods producing JG firms is suspended however. As a result of currently missing explicitly modeling of household needs (e.g. in terms of housing, medical needs, food) the model does not have a notion of unmet subsistence consumption on a per-capita level yet. Due to how consumption is generally structured in this first pass model

⁷Using a GDP ratio instead of an exogenously guessed amount keeps comparisons between runs meaningful even when different randomized starting conditions or productivity settings set the models economy to different absolute price levels.

implementing this feature would require a bigger model refactor. Without this information a Social-Gap-Index (SGI), and by extension a mission driven investment path is not definable for JG firms outside the energy sector. Contrastingly, the energy sector has a clearly assignable goal for JG firms in driving out fossil fuel use for electricity production and in industrial use.

Mission-driven investment algorithm Both JG energy firms use the same investment pipeline as private firms (L = stochastic lead time, ℓ = installation lag). The trigger that fills the pipeline is determined by policy rather than profit seeking behavior.

(A) Electricity JG firm – net-zero pathway

Based on the system-wide fossil-capacity share $s_t = K_t^{\text{fossil}}/K_t^{\text{tot}}$, a linear benchmark path $\hat{s}_t = \max\{0, s_0 - t s_0/N\}$ is determined with s_0 determined at initialization and a default of $N = 60$ monthly steps (= 5 years). Whenever $s_t > \hat{s}_t$ the required incremental clean capacity is:

$$\Delta K_t^{\text{clean}} = \max\left\{0, \frac{s_t - \hat{s}_t}{\hat{s}_t} K_t^{\text{tot}}\right\}$$

(B) Hydrogen JG firm – green-fuel pathway

Let h_t be the share of hydrogen in final energy delivered to all firms and households.⁸ The roadmap prescribes a linear lift from an initial $h_0 \simeq 0$ to an endpoint $\hat{h}_N = 0.20$ (20 %) over the same horizon $N = 60$:

$$\hat{h}_t = \min\{\hat{h}_N, t \hat{h}_N/N\}.$$

If the realized share satisfies $h_t < \hat{h}_t$ the hydrogen JG firm orders

$$\Delta K_t^{\text{H}_2} = \max\left\{0, \frac{\hat{h}_t - h_t}{1 - h_t} K_t^{\text{tot, H carrier}}\right\},$$

where $K_t^{\text{tot, H carrier}}$ is the aggregate capacity of all hydrogen-producing technologies.

(C) Kit sizing and booking

For either trigger the order size is

$$n^* = \min\left\{n^{\max}, \left\lceil \Delta K_t / \kappa_{\text{kit}} \right\rceil\right\}, \quad n^{\max} = 100,$$

with κ_{kit} the name-plate capacity of one kit. Every kit bought by the electricity JG firm is tagged with the lowest-LCOE technology in its recipe list, so the public build-out automatically tilts towards wind and solar.

The Treasury injects the necessary cash $n^* \pi_{\text{kit}}$ into the firm’s deposit, logs a JG_CAPEX event, and places the kits in the `investment_pipeline` complete with their sampled (L, ℓ) timestamps. Wage top-ups are executed by the same helper.

With this restricted but fully endogenous rule set the simulation still captures the main macro-financial feedbacks of an employer-of-last-resort programme—labor absorption, fiscal flows, and mission-oriented capital formation—while postponing the more data-hungry SGI to a future version.

Pricing rule Because JG firms are mission rather than profit-maximizing entities they adopt a stripped-down cost-plus formula that is re-evaluated every step after wages and input prices have

⁸Final energy is proxied by the sum of electricity, direct fossil fuels and hydrogen consumed in production and household recipes. The simplification ignores endogenous efficiency changes but keeps the metric inexpensive to compute.

been updated:

$$p_t^{\text{JG}} = (c_t^{\text{lab}} + c_t^{\text{mat}})(1 + \mu^{\text{JG}}), \quad \mu^{\text{JG}} \in [0.01, 0.02].$$

Here $c_t^{\text{lab}} = w_t^{\text{JG}} \ell$ is the labor cost per unit (ℓ the recipe’s labor coefficient) and $c_t^{\text{mat}} = \sum_i \alpha_i p_{i,t}$ is the contemporaneous material bill (α_i input coefficients, $p_{i,t}$ the median market price for good i). The markup μ^{JG} is configurable (default 1%). Crucially, JG firms skip the competitor-pricing branch that private firms use. Prices follow costs only, allowing the public option to undercut market prices while remaining fiscally neutral.

4 Calibration, Scaling & Model Setup

The calibration process translates an abstract agent-based structure into a fully initialized economic environment. Each step of the pipeline is designed to enforce key macroeconomic constraints such as feasibility, balanced growth, and labor consistency. Inspired by stock–flow consistent modeling principles and input–output analysis, the pipeline ensures that random draws of technology and preferences result in a structurally coherent economy. The result is a simulation-ready economy whose internal consistency is validated *ex ante* through strict guard-rails and Monte Carlo-based seed selection.

4.1 Workflow overview

- 1: Catalog build** Randomly generates the tiered set of goods, energy carriers and production recipes while enforcing Leontief feasibility.
- 2: τ -normalization** Uniformly rescales material coefficients so that the maximum column-sum of the Leontief matrix equals the target budget τ^* .
- 3: Growth-rate tuning** A second scalar brings the dominant eigenvalue $\lambda_{\max}(A)$ into a user-defined band, fixing a von-Neumann growth rate von Neumann [1945], ensuring the calibrated economy expands at a plausible and stable pace under the Leontief input–output structure.
- 4: Forward price cascade** Cost-floor prices are propagated through the input graph to seed competitive mark-ups.
- 5: Utility calibration α** An analytic formula sets the discretionary factor so that equilibrium GDP hits a multiple m of subsistence GDP.
- 6: Employment & labor-multiplier** A Newton loop on all labor coefficients targets an 80–90 % pre-shock employment share.
- 7: Infrastructure share** Service-intensity s_i is calibrated until planned spending matches an $\approx 15\%$ GDP share.
- 8: Checks and seed selection** Validates the calibrated economy against guard-rails.

4.2 Goods catalog & recipe generation

The scaling pipeline begins by calling a master builder function, which constructs a self-contained micro-economy in terms of goods and available recipes in one shot. A reproducible RNG seed fixes the draw of coefficients, utility weights and energy intensities for n consumer goods, while helpers append all special case goods. The result is a list $\mathcal{G}$ of `GoodType` records that acts as the single source of truth for all downstream modules, including a Leontief scaler and a seed hunter. The

deterministic mapper assigns tiers to goods and because material inputs are later sampled strictly from lower tiers, the procedure guarantees an acyclic input–output graph and prevents inadvertent self-feeding cycles that would make the Leontief inverse ill-defined.

τ -normalization Immediately after the raw draw each column $(a_{\bullet j}, \ell_j, m_j^{\text{mach}}, m_j^{\text{maint}})$ passes through a normalizer, which rescales material and energy inputs only until the Leontief feasibility constraint

$$\sum_i a_{ij} + \ell_j + m_j^{\text{mach}} + m_j^{\text{maint}} \leq \tau \quad (\tau=3.0 \text{ by default})$$

is satisfied. Labor and capital coefficients therefore preserve the economic structure of the original random draw and can later be shifted globally via the labor multiplier in Stage 5 without breaking feasibility.

Explicit energy technologies Before the catalog is handed to the scaler, two families of physically grounded energy recipes are appended:

- *Power generation* five technologies, coal, oil, gas, wind, solar are created. Fossil fuel coefficients are solved from the energy-balance identity $m_{\text{fuel}} = 3.6/(\eta_{\text{th}} \cdot \text{LHV})$ and each recipe carries metadata for kit capacity, labor burden, maintenance rates and, for renewables, a capacity factor distribution $(\bar{c}, \sigma_c)$.
- *Hydrogen electrolysis* stylized alkaline, PEM and SOEC variants with 40–60kWh/kg conversion factors and negligible direct CO₂ intensity are added.

The resulting triple $(\mathcal{G}, \mathcal{R}, T)$ —goods, fully normalized recipes, and a tier map—is the sole input to Stage 1 of the calibration pipeline, where global scaling begins.

4.3 Seven-stage one-shot scaler

The scaling routine converts the raw $(\mathcal{G}, \mathcal{R}, T)$ triple into a calibrated, internally consistent macro-economy with one single pass over the data. Every intermediate artefact is overwritten in place so the entire initialization can be reproduced from the random seed alone.

Notation $A \in \mathbb{R}^{n \times n}$ collects only material and energy rows (labor and machinery sit in two padding rows that are ignored unless stated otherwise); $p \in \mathbb{R}^n$ is the price vector, $L = (I - A)^{-1}$ the Leontief inverse, q_{sub} the per-household subsistence bundle, w the utility weights, $m = 3.5$ the target GDP multiple over subsistence, and $S_{\text{target}} = 0.15$ the desired infrastructure share of GDP. All steps operate with monthly units so one simulation step corresponds to 160 labor hours. Three global scalars are allowed to move freely within the routine to tune the system:

$$s_{\text{mat}}, \quad \alpha, \quad SI$$

Every other model parameter is fixed ex-ante, so the entire state of the economy is a pure function of a chosen RNG seed. All variables are expressed in monthly units (one step = 160 labor hours).

Stage 1 — τ -normalization At catalog–build time each recipe already satisfies the per-column constraint $\sum_i a_{ij} + \ell_j + m_j^{\text{mach}} + m_j^{\text{maint}} \leq \tau$ with a local budget of $\tau = 3$ (cf. Section 4.2). Once the catalog of recipes are merged into a single Leontief matrix, however, the largest material column sum $C_{\text{max}} = \max_j \sum_i a_{ij}$ can overshoot the initial budget, depending on the seed used. Running the economy with such heterogeneous material intensities would create wildly different von-Neumann growth rates and, in some seeds, an ill-conditioned $(I - A)$ inverse. Therefore a global material budget is imposed $\tau^* = 12$ that acts as a soft cap on technological density while still leaving room for capital-intensive industries. The scalar s_{mat} may move at this stage multiplying all material and energy coefficients, leaving labor and machinery untouched:

$$a_{ij}^{\text{new}} = s_{\text{mat}} a_{ij}^{\text{old}}, \quad \ell_j^{\text{new}} = \ell_j^{\text{old}}, \quad m_j^{\text{new}} = m_j^{\text{old}}$$

Keeping labor unscaled is crucial: it preserves the Hawkins–Simon sign pattern Hawkins and Simon [1949] and guarantees that wages retain leverage over the IO system.

The scaler locates s_{mat} with a standard bisection procedure. It begins with the interval $[0, 10^3]$ and, at each trial value s , rebuilds the materials-only block of the Leontief matrix after multiplying all material and energy coefficients by s . From that matrix it computes the largest column sum $C_{\text{max}}(s) = \max_j \sum_i a_{ij}(s)$. If $C_{\text{max}}(s) > \tau^*$ the factor s is too large and the upper half of the bracket is discarded, if it falls below the target, the lower half is removed. The interval therefore shrinks by one half each iteration until the stopping criterion $|C_{\text{max}}(s) - \tau^*| < 10^{-3}$ is met or twenty iterations have elapsed. Because labor and machinery rows are never touched, any Hawkins–Simon feasibility present before the search remains intact. The search itself, however, does not enforce that condition. A dedicated check in Stage 7 will do so after growth-rate adjustment.

Once s_{mat} is fixed the scaler records (i) the mean labor coefficient $\bar{L}$ and the mean material coefficient $\bar{M}$; (ii) the final C_{max} to verify the target; and (iii) floors all labor coefficients below 0.35 (0.50 for tier-0 raw-material recipes) to prevent ‘ghost firms’ employing fewer than one worker. The rescaled matrix $A^{(1)} = s_{\text{mat}} A$ is then passed to Stage 2.

Stage 2 — growth-band scaling The Leontief block that emerges from Stage 1 is still free to generate implausibly high or low balanced-growth rates. The speed of that balanced path is fixed by the dominant eigenvalue of the material block:

$$\lambda_0 = \lambda_{\text{max}}(A^{(1)}), \quad \rho_0 = \frac{1}{\lambda_0} - 1,$$

with $A^{(1)} = s_{\text{mat}} A$ from Stage 1. For a small demo economy we want the von-Neumann rate ρ_0 in a reasonable interval. In terms of the eigenvalue this means λ_0 must lie in $[\lambda_{\text{lo}}, \lambda_{\text{hi}}] = [0.7, 0.9]$.

Because every input coefficient is linear, one global multiplier s_λ is enough to move the whole spectrum:

$$\lambda_{\text{max}}(s_\lambda A^{(1)}) = s_\lambda \lambda_{\text{max}}(A^{(1)}) = s_\lambda \lambda_0.$$

Hence we can pick s_λ in closed form:

$$s_\lambda = \begin{cases} \frac{\lambda_{\text{hi}}}{\lambda_0}, & \lambda_0 > \lambda_{\text{hi}} \quad (\text{economy too 'dense'}); \\ \min\left\{\frac{\lambda_{\text{lo}}}{\lambda_0}, \frac{\tau^*}{C_{\text{max}}(A^{(1)})}\right\}, & \lambda_0 < \lambda_{\text{lo}} \quad (\text{economy too 'sparse'}); \\ 1, & \text{otherwise.} \end{cases}$$

Here $C_{\text{max}}(A^{(1)}) = \max_j \sum_i a_{ij}^{(1)}$ is the largest column sum after Stage 1. The $\min\{\cdot\}$ makes sure we never break the global feasibility ceiling $C_{\text{max}} \leq \tau^*$ when we inflate the matrix.

No iterative search is needed: a single eigen-decomposition, a standard NUMPY call, gives λ_0 , after which the suitable s_λ follows directly. Applying it to all material and energy inputs produces

$$A^{(2)} = s_\lambda A^{(1)}, \quad \lambda_{\text{final}} = s_\lambda \lambda_0, \quad \rho_{\text{final}} = \frac{1}{\lambda_{\text{final}}} - 1.$$

Because s_λ is capped as shown above, the Hawkins–Simon condition $\det(I - A^{(2)}) > 0$ automatically stays true. The rescaled matrix $A^{(2)}$ is passed unchanged to the price-building routine in Stage 3.

Stage 3 — cascading prices Base prices are built once, forward through the topological order implied by the tier map. The cost floor for output good g is $c_g^{\text{lab}} + c_g^{\text{mat}}$ (where labor cost uses the exogenous minimum wage $w = 12\text{€}/\text{h}$). A uniform markup μ_g is applied and a universal penny-floor avoids numerical zeroes:

$$p_g = \max\left\{(c_g^{\text{lab}} + c_g^{\text{mat}})(1 + \mu_g), 0.01\right\}.$$

Stage 4 — discretionary factor α Benchmark household demand is split into two fixed-proportion bundles

a) a subsistence bundle $q_{\text{sub}} \in \mathbb{R}^n$;

b) a discretionary bundle whose composition is given by a non-negative weight vector $w \in \mathbb{R}^n$.

Both bundles are specified on a per-household basis. With $H = 200$ test households the resulting final-demand vectors are

$$d_{\text{sub}} = H q_{\text{sub}}, \quad d_{\text{disc}} = H w.$$

Given the - and growth-scaled input matrix $A^{(2)}$ from Stage 2, the gross-output vectors that satisfy either bundle are obtained via the Leontief inverse $L = (I - A^{(2)})^{-1}$:

$$x_{\text{sub}} = L d_{\text{sub}}, \quad x_{\text{disc}} = L d_{\text{disc}}.$$

The discretionary bundle is multiplied by a single scalar α , so benchmark nominal GDP is

$$\text{GDP}(\alpha) = p^\top (x_{\text{sub}} + \alpha x_{\text{disc}}),$$

with p the price vector from Stage 3. The model requires total GDP to be $m = 3.5$ times the pure-subsistence level $\text{GDP}_{\text{sub}} = p^\top x_{\text{sub}}$. Imposing $\text{GDP}(\alpha) = m \text{GDP}_{\text{sub}}$ and solving for α yields a closed-form expression:

$$p^\top (x_{\text{sub}} + \alpha x_{\text{disc}}) = m p^\top x_{\text{sub}} \implies \boxed{\alpha = \frac{(m-1) p^\top x_{\text{sub}}}{p^\top x_{\text{disc}}}}.$$

The numerator represents the extra income that needs to be created, the denominator is the cost of one unit of the discretionary basket. The ratio can be interpreted as the 'number of discretionary bundles per household'. Random seeds occasionally make the discretionary goods extremely cheap, inflating α and through it the entire scale of the economy. To avoid such pathological cases a smooth limiter is applied

$$\alpha \leftarrow 150 \tanh(\alpha/150),$$

which keeps real incomes in a plausible range without introducing a hard discontinuity. α is stored in the global parameter dictionary and passed forward to Stage 5, where labor demand is reevaluated with the full household budget in place.

Stage 5 — labor multiplier At this point the material matrix $A^{(2)}$, the price vector p and the discretionary factor α are fixed. Running a planning sub-routine once with best-technology recipes, i. e. those with the lowest labor-hour requirement per good, yields a gross-output vector $x^{\text{best}} \in \mathbb{R}^n$. The sub-routines task is to translate final demand into (i) gross-output targets, (ii) the number of workers, and (iii) the machine hours each sector will need. The routine is deterministic given the current recipe catalog and parameter set. During calibration it is invoked up to a few dozen times with updated labor coefficients or infrastructure demand. For every good $g \in \mathcal{G}$

the catalogue usually contains several recipes. To bracket a plausible labour range the planner constructs two Leontief blocks:

A^{best} : pick the *cheapest* recipe in labor hours for each g ,

A^{worst} : pick the *most* labor-intensive recipe for each g .

All subsequent best / worst figures are computed with these two matrices.

The power sector is the only place where multiple parallel recipes have to be considered to result in a varied electricity mix. Instead of choosing a single technology the planner folds them into one composite column.

- a) Draw a random electricity mix $S_{\text{fossil}}, S_{\text{wind}}, S_{\text{solar}}$ with $0.40 \leq S_{\text{fossil}} \leq 0.70$ and a minimum of 5% per individual source. Implemented by a helper that: (i) uniformly samples the total fossil share, (ii) splits it across Coal/Oil/Gas via a Dirichlet draw, (iii) allocate the residual to wind and solar, again with a Dirichlet draw.
- b) Replace the “Electricity” column of A^{best} and A^{worst} by the *weighted* sum of the individual recipe columns, e. g. $a_{ij}^{(\text{power})} = S_{\text{coal}} a_{ij}^{\text{coal}} + S_{\text{gas}} a_{ij}^{\text{gas}} + \dots$.

A record of those weights is shipped with the final scaling plan so that simulated emissions can be reconciled with the calibration.

With a Leontief block at hand and a household demand vector $d \in \mathbb{R}^n$ (subsistence + discretionary + infra) the gross output that exactly meets final demand is

$$x = (I - A)^{-1}d.$$

The computation is carried out twice, once with A^{best} and once with A^{worst} .

Given x the total labor hours read

$$H_{\text{tot}} = \sum_g x_g \ell_g, \quad \implies \quad W = \frac{H_{\text{tot}}}{160},$$

where ℓ_g are the current labor coefficients. The same summation with machinery coefficients yields machine-hour requirements.

Finally the planner splits W across firms. A log-normal draw adds mild size dispersion so that not every firm is identical at the outset.

All quantities produced by this subroutine are strictly algebraic functions of the recipe catalog and the current set of scalars $s_{\text{mat}}, s_{\lambda}, \alpha, \text{SI}$. Re-running it after one of those scalars changed therefore gives an immediate, globally consistent update without the need for iterative market clearing – a key reason why the entire seven-stage scaler stays fast enough to be embedded inside the seed-hunter.

Total monthly labor hours implied by that plan are

$$H_{\text{tot}} = \sum_{g \in \mathcal{G}} x_g^{\text{best}} \ell_g^{(2)},$$

where $\ell_g^{(2)}$ are the labor coefficients after Stages 1–4. Dividing by the length of one model period ($h_{\text{step}} = 160$ h) gives the planned head-count

$$W_{\text{plan}} = \frac{H_{\text{tot}}}{h_{\text{step}}}.$$

With $H = 200$ households the corresponding employment rate is simply

$$\eta = \frac{W_{\text{plan}}}{H}.$$

Calibration aims for an 80% pre-shock employment share, $\eta^* = 0.80$. Because every labor coefficient enters the total linearly, the gap to that target can be closed with a single global multiplier:

$$f = \frac{\eta^*}{\eta}, \quad \ell_g^{\text{new}} = f \ell_g^{\text{old}} \quad \forall g \in \mathcal{G}.$$

Scaling all ℓ_g by the same factor multiplies total hours, and therefore the head-count.

In practice a light damping is applied to avoid oscillations when the machine-capacity side of the economy becomes binding. At iteration k

$$f_k = 0.5 \frac{\eta^*}{\eta_k}, \quad \ell_g^{(k+1)} = f_k \ell_g^{(k)},$$

followed by a new planning routine call that yields the updated η_{k+1} . The procedure stops once $|\eta_{k+1} - \eta^*| < 0.01$ or after a hard cap of 20 rounds. After every update labor coefficients are floored at 0.35), and 0.50 for tier-0 raw goods to prevent “ghost firms” that would employ less than a full worker. If the planning routine reports machine-hour utilization below 80% the loop terminates early: further labor scaling would not translate into additional employment once capital becomes the bottleneck.

The final coefficient set $\ell_g^{(3)}$ achieves an employment rate of $\eta \simeq \eta^*$ while preserving Hawkins–Simon feasibility. These coefficients, together with the updated output plan, feed into the infrastructure calibration of Stage 6.

Stage 6 — infrastructure calibration. The final macro dial is the service-intensity SI of the public-infrastructure sector, measured in ‘infra-hours’ demanded

$$\text{SI} \left[\frac{\text{infra_service units}}{\text{household}} + \frac{\text{infra_service units}}{\text{unit gross output}} \right].$$

For any tentative value of SI the planning routine returns

$$Y = \text{GDP}(\text{SI}), \quad I = p_{\text{infra}} D_{\text{infra}}(\text{SI}),$$

where p_{infra} is the cost-floor price from Stage 3 and D_{infra} is the sum of household and firm demand for infrastructure. Because D_{infra} is linear in SI the spending share

$$S_{\text{actual}} = \frac{I}{Y}$$

scales proportionally. To hit the policy target $S_{\text{target}} = 15\%$ we therefore need nothing more sophisticated than one-dimensional proportional control:

$$\text{SI}_{k+1} = \text{SI}_k \frac{S_{\text{target}}}{S_{\text{actual},k}}, \quad k = 0, 1, 2, \dots$$

Starting from a safe guess ($\text{SI}_0 = 0.5$ infra-hours per household and 0.07 per unit output in the code) the loop is executed four times, which suffices for every seed tested to achieve

$$|S_{\text{actual},k} - S_{\text{target}}| < 10^{-3}.$$

Because the update is exact for a linear mapping the first iteration is already very close, but rounding effects in the Leontief inverse and in the price cascade introduce a small residual. The error shrinks by roughly an order of magnitude each pass, so after four rounds the target share is met.

The calibrated intensity SI^* is written back into the global parameter set. With labour, income and infrastructure now all tuned, the economy is fully specified for the final sanity checks in Stage 7.

Stage 7 — checks and seed-selection The scaler finishes by validating the entire calibrated economy against a set of macro guard-rails. If the current random seed fails one of the tests, the whole pipeline is rerun with a fresh seed until a fully admissible configuration is found.

Guard-rails. Let ρ be the von-Neumann growth rate from Stage 2, S_{ren} the ex-ante renewable share in the electricity mix (wind + solar), S_{infra} the infrastructure spending fraction from Stage 6, and $\eta_{\text{best}}, \eta_{\text{worst}}$ the employment ratios of the optimistic and pessimistic recipe sets, respectively. The candidate economy is kept if

$$\begin{aligned} \text{(a)} \quad & 0.25 \leq \rho \leq 0.45, & \text{(b)} \quad & 0.30 \leq S_{\text{ren}} \leq 0.50, \\ \text{(c)} \quad & 0.12 \leq S_{\text{infra}} \leq 0.18, & \text{(d)} \quad & 0.80 \leq \eta_{\text{best}} \leq 0.95, \\ \text{(e)} \quad & 0.80 \leq \eta_{\text{worst}} \leq 1.00, & \text{(f)} \quad & C_{\text{max}}(A^{(2)}) \leq 1 \quad (\text{Hawkins-Simon}), \\ \text{(g)} \quad & p_g > 0 \quad \forall g \in \mathcal{G}. \end{aligned}$$

The helper script encapsulates this logic. Starting from a random seed it:

- a) runs the seven-stage scaler,
- b) records which criterion caused a rejection, and
- c) retries with a fresh seed until the first fully admissible economy is produced (default limit: 1 000 trials).

Summary statistics for a diagnostic sweep of 1000 seeds read:

Metric	Mean	Std	Med.	Q_{25}	Q_{75}	CV %
von-Neumann rate ρ	0.330	0.066	0.333	0.290	0.373	19.9
Renewable share S_{ren}	0.448	0.076	0.443	0.392	0.513	17.0
Infra share S_{infra}	0.146	0.005	0.146	0.146	0.147	3.3
Employment (worst) η_{worst}	0.944	0.091	0.950	0.925	0.970	9.6
Discretionary factor α	128	10.5	128	123	134	8.3
Service intensity SI	21.49	2.85	21.46	20.08	23.17	13.3

Table 3: Distribution of key macro indicators across 1 000 random seeds before applying the guard-rails. CV = coefficient of variation.

The table shows that most draws sit in the desired bands after calibration. The renewable share is the noisiest dial (CV $\approx 17\%$), low renewable shares and von-Neumann rates out of the target band are usually the main reason a seed gets rejected. In a typical run the seed hunter needs only a few attempts to land a valid economy. The first seed that passes all tests is returned as a **ScalingPlan** structure, returned by the planning sub-routine, containing all needed information for simulation setup.

4.4 Simulation initialization

With a valid **ScalingPlan** in hand the model enters the setup phase. The routine starts by loading a configuration file. At the same time every parameter that the one-shot scaler determined endogenously is merged into the same dictionary, guaranteeing that one coherent parameter set governs both calibration and simulation.

The seed-hunter supplies an RNG seed that already satisfies the macro guard-rails of Stage 7. Both **numpy** and the built-in **random** module are re-seeded with this value such that *all* stochastic draws made during initialization and run-time become reproducible. This the same world can

also run through different scenario configurations, while being comparable to the same worlds baseline scenario outcomes. The scalers output is then unpacked: the list of goods and recipes, tier assignments, the complete price dictionary as well as firm and worker targets. Every `GoodType` is updated so that its `base_price` equals the cost-floor price computed by the scaler.

Agents are initiated next according to configuration parameters. The financial core is built by one central bank, one government and a number of commercial banks. Households are generated and each is given a Dirichlet-drawn preference vector over all consumer goods whose utility weight is positive. Firms are instantiated exactly as prescribed by the scaling plan. All agents get assigned a static reference bank for accounting purposes. For every good g the plan lists how many firms should exist, how many workers each should employ and how much capacity should be allocated to each firm. That information is copied one-to-one. Regular producers inherit a log-normal markup draw ($\mu = \ln 1.2$, $\sigma = 0.3$, 1 % floor).

Recipes are randomly attached to firms and inventories are then primed. Every producer starts with one step’s worth of its own output so that households can consume their target bundle immediately, and three steps’ worth of all intermediate inputs that occur in its recipes. Fossil generators receive a fuel buffer and renewable plants get maintenance kits according to their expected utilization.

A fiscal injection then seeds the monetary circuit. The government credits deposits equivalent to 1.5 months of payroll to every firm and three months to every household, financed by the issuance of bonds. If the active scenario flag requests a Job-Guarantee, the government spawns the corresponding public-service firms at a wage $w_{JG} = 1.05 w_{\min}$.

At this point the entire agent set is live and handled by the phases outlined initially. From here on every event in the simulation is endogenous.

5 Conclusion and Outlook

The thesis set out to determine whether a stock–flow–consistent, agent-based macroeconomic model can be extended to encompass explicit energy flows, greenhouse-gas accounting and heterodox policy instruments. The resulting prototype demonstrates four central achievements. First, it confirms that monetary balance-sheet consistency, physical throughput constraints and behavioral rules can live within one executable framework and be generally scalable. Second, by modeling households, firms, financial institutions and government as distinct yet interacting classes which can naturally split into finer-grained agent types through inheritance mechanics, the code base remains open to incremental increases in heterogeneity and path dependence. Third, the built-in carbon-pricing switch and the mission-oriented Job-Guarantee show that market instruments and direct public provisioning can coexist and be compared inside the same architecture. Fourth, the automatic seven-stage scaling routine turns a mostly stochastic catalog of recipes into a balanced initial condition that meets macro-targets by construction, converting what is a laborious calibration exercise into a reproducible one-shot procedure, while keeping the catalog of goods extensible.

Even so, the present version must be treated as a starting point for a simulation platform consisting of integrated sub-systems, not a finished analytical tool. Most modules require further attention and development time to enable more detailed, agent specific and realistic behavior. The carbon tax model is only a proxy for the European Emissions Trading System: Allowance banking, the Market Stability Reserve and sectoral coverage differences have not yet been modeled. Household behavior is reduced to fixed budget shares, firms lack cost learning and strategic pricing, and the labor market omits skills, bargaining and demographic flows. The economy is represented as a single region, so transport, land use and grid congestion are absent. Interbank and asset markets are missing. Resources are still generally treated as infinite and there are no demographic dynamics.

Performance is limited by the use of pure Python and a migration to a more performative language as well as the adoption of an entity component system will be needed at some point in the future. A phased roadmap follows naturally. In the short term the consumption block should adopt a constant elasticity of substitution formulation and explicit modeling of basic needs. Labor should gain skill types, vacancy posting and educational requirements. Demographic developments should be covered. Over the medium term the stand-in carbon tax should give way to a dynamic ETS with auctions, allowance banking and MSR feedback, possibly in the form of another configurable scenario. The power sector needs to be modeled in more detail in terms of plant specific characteristics. In the long term the one-country closure ought to evolve into a multi-country, multi-currency setup with cross-border trade, capital flows and differentiated sovereign monetary regimes, enabling comparative analysis of green industrial strategies. A notion of geographic space may be implemented to capture spatial dynamics and constraints. Additionally, developing a sensible user interface will make sense so parameters, starting conditions and behavioral routines may be adjusted more easily for experiments.

No quantitative results are presented here because any headline figures would imply a level of precision the prototype does not possess at the time of writing. Core behavioral routines are provisional, many parameters are still placeholders by necessity, and the numerical implementation has not undergone a full stability, convergence or sensitivity audit. The system itself needs to undergo more rigorous evaluation since there are signs hidden bugs, that do not break model execution but possibly alter results, could still be present in the code base as a result of quick iteration over thousands of lines of code. The value of this thesis therefore lies in its architecture: it provides a logically coherent, physically anchored and monetarily consistent scaffold on which richer behavioral, institutional and spatial detail can now be layered. Turning this scaffold into a policy-grade laboratory that is able to highlight interactions between larger systems is a longer-term task, that will require iterative refinement, empirical anchoring and computational strengthening. The conceptual foundations are in place.

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